SIAM J. MATRIX ANAL. APPL.
Vol. 47, No. 2, pp. 649–672

EFFICIENT ESTIMATE FOR THE OPTIMAL BACKWARD ERROR OF THE MULTIDIMENSIONAL TOTAL LEAST SQUARES*

JIALI SHAN† AND YIMIN WEI‡

Abstract. In this paper, we originally derive the explicit expression for the optimal backward error of the multidimensional total least squares (mTLS) problem, with both full-rank and rank-deficient computed solutions considered. We propose the necessary and sufficient conditions for mTLS solutions, constructively characterizing the feasible set of the targeted perturbations. These results reveal deep connections between least squares (LS) and total least squares (TLS) and highlight the inherent difference in their problem structures. Theoretical analysis is still computationally prohibitive for large-scale applications. To address this issue, we propose computable estimates with tight bounds and further introduce randomized algorithms based on sketching techniques. Numerical experiments illustrate the effectiveness and the efficiency of the algorithms, offering a practical tool for iterative refinement. This work facilitates the backward stability analysis for the TLS problems.

Key words. total least squares, backward error analysis, randomized algorithm, sketching

MSC codes. 65F08, 65F20, 65G50, 68W20

DOI. 10.1137/25M1788312

See reproducibility of computational results at end of the article.

1. Introduction. Total least squares problems with multiple right-hand sides, also called multidimensional total least squares (mTLS), arise in many applications, such as data fitting, image processing, system identification, etc. [22, 32, 42]. The mTLS problem finds the best fit $X \in \mathbb{R}^{n\times d}$ to an overdetermined system $AX \approx B$, where $A \in \mathbb{R}^{m\times n}$ and $B \in \mathbb{R}^{m\times d}$ $(m \gg n > d)$, formally given by

$$\min_{\Delta A, \Delta B} \|[\Delta A, \Delta B]\|_{\mathrm{F}}, \ \text{s.t.}\ (A+\Delta A)X = B + \Delta B. \tag{1.1}$$

Let $[A,B] = U\Sigma V^\top$ be the singular value decomposition (SVD) of $[A,B]$, where $\Sigma = \mathrm{diag}(\sigma_i)$ with $\{\sigma_i\}_{i=1}^{n+d}$ arranged in nonincreasing order, and V_{22} denotes the bottom-right d-by-d block of V. Much research has explored the solvability of mTLS systems [23, 24, 25, 26, 27, 31, 47, 48]. Hereafter, we assume the *genericity condition*

$$\sigma_n([A,B]) > \sigma_{n+1}([A,B]) \text{ and } V_{22} \text{ nonsingular}, \tag{1.2}$$

to ensure the existence and uniqueness of the mTLS solution [14, 41].

For any numerical problem, one of the most significant topics people are concerned about is the backward perturbation analysis because it is vital for the theoretical

*Received by the editors August 14, 2025; accepted for publication (in revised form) by M. Tůma January 8, 2026; published electronically May 27, 2026.
https://doi.org/10.1137/25M1788312
Funding: The first author was supported by the National Natural Science Foundation of China (Youth Student Basic Research Program) under grants 123B1008 and 12271108. The second author was supported by the National Natural Science Foundation of China under grant U24A2001 and by the Science and Technology Commission of Shanghai Municipality under grant 23JC1400501.
†School of Mathematical Sciences, Fudan University, Shanghai 200433, People's Republic of China (jlshan25@m.fudan.edu.cn).
‡Corresponding author. School of Mathematical Sciences and Key Laboratory of Mathematics for Nonlinear Sciences, Fudan University, Shanghai 200433, People's Republic of China (ymwei@fudan.edu.cn).

stability and sometimes the stopping criteria of the iterative algorithms. In this paper, we focus on the analysis of (1.1) and estimate the optimal backward error with efficient algorithms.

Given a computed solution $\widetilde{X} \in \mathbb{R}^{n\times d}$, the weighted backward error of (1.1) is defined with the weighting parameter θ by

$$\min\Big\{ \|[E,\theta F]\|_{\mathrm{F}} : \widetilde{X} \text{ is the true solution to a nearby mTLS system} \\ \min_{\Delta A,\Delta B} \|[\Delta A,\Delta B]\|_{\mathrm{F}},\ \text{s.t. } [(A+E)+\Delta A]\widetilde{X} = (B+F)+\Delta B \Big\}. \tag{1.3}$$

The parallel problem in the least squares (LS) setting, which only assumes noise from the right-hand side, has been extensively studied over the past decades. The most influential works on this topic include the explicit theoretical backward error expression [38, 43], the corresponding effective estimate with a tight bound [15, 17, 18, 30], and its applications to the design space of stable LS solvers [12].

However, such theories for total least squares (TLS) are still underdeveloped despite the foundational work dating back to the last century [40, 41]. Also, the connections between the LS and the TLS should be further clarified for potentially leveraging advances in one to inform the other. Therefore, our goals are to

1. *Theoretically* establish the backward perturbation theory for the general mTLS problems, elucidating the relations between the LS and TLS structures;
2. *Practically* derive tight bounds for the estimation of the optimal backward error and develop efficient algorithms for its computation.

Inspired by the decades-long evolution of the LS methodology, our research follows a similar trajectory and aims to provide a comprehensive answer to (1.3).

1.1. Notation and preliminaries. $\|\cdot\|_{\mathrm{F}}$ denotes the Frobenius norm, while $\|\cdot\|_2$ or $\|\cdot\|$ denotes the spectral norm of a matrix or the Euclidean norm of a vector. $\|\cdot\|_*$ denotes the nuclear norm, which is the sum of the singular values of a matrix. For a square matrix A, the trace $\mathrm{tr}(A)$ is defined as the sum of its diagonal entries. I_n denotes an n-by-n identity matrix. For $P\in\mathbb{R}^{m\times n}, Q\in\mathbb{R}^{p\times q}$, their Kronecker product is defined by $P\otimes Q = (p_{ij}Q)\in\mathbb{R}^{mp\times nq}$. Assuming Q invertible and Q_1, Q_2 of the same size, there are beautiful identities that

$$(Q\otimes I_r)^\top = Q^\top\otimes I_r,\quad (Q\otimes I_r)^{-1} = Q^{-1}\otimes I_r,\quad (Q_1\otimes I_r)(Q_2\otimes I_r) = (Q_1Q_2)\otimes I_r. \tag{1.4}$$

$\mathrm{Vec}(T)$ is a vector written by stacking the columns of the matrix T. For all compatible A, B, X and any positive definite $S\in\mathbb{R}^{m\times m}$, the following strong relations hold:

$$\mathrm{Vec}(AXB) = (B^\top\otimes A)\mathrm{Vec}(X),\quad \mathrm{tr}(A^\top B) = \mathrm{Vec}(A)^\top\mathrm{Vec}(B); \tag{1.5}$$

$$\mathrm{Vec}(M)^\top(S\otimes I_m)\mathrm{Vec}(M) = \mathrm{tr}(SM^\top M). \tag{1.6}$$

For further reference, we recommend [28] for a detailed presentation of properties of the Kronecker product and the Vec operator. $T^\dagger$ denotes the Moore–Penrose inverse [45] of the matrix T, while that of any vector $v\in\mathbb{R}^n$ is denoted by

$$v^\dagger \equiv \begin{cases} 0 & \text{if } v=0,\\ v^\top/\|v\|^2 & \text{if } v\neq 0.\end{cases}$$

If T is of full column rank, then $T^\dagger = (T^\top T)^{-1}T^\top$ and $T^\dagger T = I$. If T is of full row rank, then $T^\dagger = T^\top(TT^\top)^{-1}$ and $TT^\dagger = I$.

1.2. Related works. In the following, we briefly review relevant existing literature on the perturbation analysis for the LS and TLS problems.

Least squares (LS) problems. In [30, 43], Waldén, Karlson, and Sun for the first time characterize the feasible sets of the perturbations in the LS problem with a single right-hand side. For the LS system $Ax \approx b$ with $A \in \mathbb{R}^{m\times n}$ and $b \in \mathbb{R}^m$, given a nonzero $\widehat{x} \in \mathbb{R}^n$ and letting $\widehat{r} = b - A\widehat{x}$, the following two sets are identical:

$$\mathcal{E} = \left\{ E \in \mathbb{R}^{m\times n} : \|b - (A+E)\widehat{x}\| = \min_{x\in\mathbb{R}^n} \|b - (A+E)x\| \right\},$$
$$\mathcal{M} = \left\{ \widehat{r}\widehat{x}^\dagger - vv^\dagger(A + \widehat{r}\widehat{x}^\dagger) + (I - vv^\dagger)Z(I - \widehat{x}\widehat{x}^\dagger) : v \in \mathbb{R}^m, Z \in \mathbb{R}^{m\times n} \right\}.$$

They further derive the optimal backward error formula that accounts for both perturbations in A and b. Following the previous notation and given the weighting parameter θ, define $\lambda_\theta = \lambda_{\min}(AA^\top - \tau\frac{\widehat{r}\widehat{r}^\top}{\|\widehat{x}\|^2})$ as the smallest eigenvalue of $(AA^\top - \tau\frac{\widehat{r}\widehat{r}^\top}{\|\widehat{x}\|^2})$ with $\tau = \theta^2\|\widehat{x}\|^2/(1+\theta^2\|\widehat{x}\|^2)$ and denote

$$\mathcal{G} = \left\{ [E, f] : \|(b+f) - (A+E)\widehat{x}\| = \min_{x\in\mathbb{R}^n} \|(b+f) - (A+E)x\| \right\}.$$

Then the weighted backward error for the LS problem is

$$\min_{[E,f]\in\mathcal{G}} \|[E, \theta f]\|_{\mathrm{F}} = \left[\tau\frac{\|\hat{r}\|^2}{\|\hat{x}\|^2} + \min\{0, \lambda_\theta\} \right]^{\frac{1}{2}}. \tag{1.7}$$

Results for the multidimensional LS (mLS) problems are extended in [37]. The optimal normwise backward error for linear least squares (LLS) problems and least squares problems with equality constraints (LSE) are shown in [9, 38].

For the sake of computational convenience, error bounds and their estimates are carefully studied. For the remainder of this paragraph, we set $\theta = 1$ for simplicity. First of all, it is recommended by Higham [21] that (1.7) can be equivalently expressed as

$$\mu(\widehat{x}) = \min\left\{ \frac{\|\widehat{r}\|}{\|\widehat{x}\|}, \sigma_{\min}[A, B] \right\}, \quad B = \frac{\|\widehat{r}\|}{\|\widehat{x}\|}\left(I - \widehat{r}\widehat{r}^\dagger\right),$$

where $\sigma_{\min}[A, B]$ is the smallest singular value of $[A, B]$. Karlson and Waldén [30] prove that calculating the error with respect to the Frobenius norm or the spectral norm only differs by a small constant factor, and then they propose a computable estimate starting from the derivation of the spectral norm formulation, namely,

$$\widetilde{\mu}(\widehat{x}) = \left\| \left(\|\widehat{x}\|^2 A^\top A + \|\widehat{r}\|^2 I\right)^{-\frac{1}{2}} A^\top\widehat{r} \right\|, \text{ with } \widetilde{\mu}(\widehat{x})/\mu(\widehat{x}) \le \tfrac{2+\sqrt{2}}{2}. \tag{1.8}$$

Gu [18] improves the result to $\|r_{\mathrm{LS}}\|/\|\widehat{r}\| \le \widetilde{\mu}(\widehat{x})/\mu(\widehat{x}) \le (1+\sqrt{5})/2$ with $r_{\mathrm{LS}} = b - Ax_{\mathrm{LS}}$ being the true LS residual. Grcar [16] concludes that $\widetilde{\mu}(\widehat{x})$ is asymptotically equal to $\mu(\widehat{x})$ as $\widehat{x}$ approximates x_{LS}. Finally, Gratton et al. [15] provide a refined analysis to show that $\widetilde{\mu} \le \mu \le \sqrt{2}\widetilde{\mu}$. Numerical tests and more discussions on the accuracy of (1.8) can be found in [17]. Approximation for the theoretical backward error of mLS problems is considered in [20].

Total least squares (TLS) problems. Given $A \in \mathbb{R}^{m\times n}$, $b \in \mathbb{R}^m$ and a scaling parameter γ, a scaled total least squares (STLS) problem [34] solves the system $Ax \approx b$ in the sense of

$$\min_{\Delta A, \Delta b} \|[\Delta A, \gamma\Delta b]\|_{\mathrm{F}}, \text{ s.t. } (A + \Delta A)x = b + \Delta b. \tag{1.9}$$

The weighted backward error for (1.9) is naturally defined with parameter θ as

$$\min_{E,f} \|[E, \theta f]\|_{\mathrm{F}}, \text{ s.t. } \widetilde{x} \text{ solves a nearby STLS problem } (A+E)\widetilde{x} \approx b+f. \tag{1.10}$$

Throughout this paper, we focus on the case where $\gamma = 1$. Chang, Paige, and Titley-Péloquin [5] establish the matrix characterizations for any compatible systems and perturbations in the LS and STLS systems. Based on these characterizations, they obtain the optimal backward error expression for the STLS problem [6]. Given a computed solution $\widetilde{x} \in \mathbb{R}^n$, let $\widetilde{r} = b - A\widetilde{x}$. Then, (1.10) is explicitly given by

$$\min_{E,f} \|[E, \theta f]\|_{\mathrm{F}}^2 = \begin{cases} \dfrac{\theta^2\|\widetilde{r}\|^2}{1+\theta^2\|\widetilde{x}\|^2} & \text{if } \lambda_{\min}(M) \geq 0, \\ \dfrac{\theta^2\|\widetilde{r}\|^2}{1+\theta^2\|\widetilde{x}\|^2} + \lambda_{\min}(M) = \sigma_{\min}(N) & \text{if } \lambda_{\min}(M) < 0, \end{cases} \tag{1.11}$$

where the two matrices N and M are

$$\begin{cases} N = \left[A(I - \widetilde{x}\widetilde{x}^\dagger), \dfrac{\theta\|\widetilde{r}\|}{\sqrt{1+\theta^2\|\widetilde{x}\|^2}}(I - \widetilde{r}\widetilde{r}^\dagger), \dfrac{\theta}{\sqrt{\theta^2\|\widetilde{x}\|^2 + \gamma^4\|\widetilde{x}\|^4}}(A\widetilde{x} + \gamma^2\|\widetilde{x}\|^2 b) \right], \\ M = A(I - \widetilde{x}\widetilde{x}^\dagger)A^\top - \dfrac{\theta^2\widetilde{r}\widetilde{r}^\top}{1+\theta^2\|\widetilde{x}\|^2} + \dfrac{\theta^2(A\widetilde{x} + \gamma^2\|\widetilde{x}\|^2 b)(A\widetilde{x} + \gamma^2\|\widetilde{x}\|^2 b)^\top}{\theta^2\|\widetilde{x}\|^2 + \gamma^4\|\widetilde{x}\|^4}. \end{cases}$$

In [36], a more concise expression is derived, theoretically supporting the design of a better solver based on LS techniques. While the main idea in [36] holds, we find some typos therein, correct them in this paper, and also provide a new proof for better understanding. The correct statement should be as follows. Let $A \in \mathbb{R}^{m\times n}, b \in \mathbb{R}^m$, and the computed solution $\widetilde{x} \in \mathbb{R}^n$. Write $\widetilde{A} = (A + b\widetilde{x}^\top)(I + \theta^{-2}\widetilde{x}\widetilde{x}^\top)^{-\frac{1}{2}}$ and $\widetilde{r} = \theta(1+\theta^2\|\widetilde{x}\|^2)^{-\frac{1}{2}}(b - A\widetilde{x})$. Denote λ_θ as the smallest eigenvalue of the matrix $\widetilde{A}\widetilde{A}^\top - \widetilde{r}\widetilde{r}^\top$. Then, the optimal backward error of a TLS is given by

$$\min_{E,f} \|[E, \theta f]\|_{\mathrm{F}}^2 = \left[\|\widetilde{r}\|^2 + \min\{0, \lambda_\theta\}\right]^{\frac{1}{2}}.$$

Please refer to Theorem 2.8 and Corollary 2.9 in the current paper.

1.3. Outline. Section 2 achieves our first goal, establishing the backward error analysis for the mTLS system. Section 3 tackles the second goal, proposing effective estimates for the previous theory with tight bounds. Fast randomized algorithms based on sketching techniques are introduced to efficiently approximate the optimal backward error. Section 4 presents various numerical experiments to validate the effectiveness and the efficiency of the proposed methods. Finally, section 5 concludes the paper and discusses future work.

2. Optimal backward error formula for mTLS.

In the following, we construct the theory for the optimal backward error in the mTLS problem. We first propose the necessary and sufficient conditions for the mTLS solutions, then characterize the feasible set of perturbations, and finally derive the explicit formula. Both full-rank and rank-deficient computed solutions are considered.

2.1. The necessary and sufficient conditions for mTLS solutions. We construct the theory for later backward error analysis.

THEOREM 2.1 (Necessary and sufficient conditions for the mTLS solutions). *Given the genericity condition* (1.2), *X is the true solution to the mTLS problem* (1.1) *if and only if the following two conditions hold:*

$$\left(A+BX^\top\right)^\top (AX-B)=\mathbf{0}, \tag{2.1}$$

$$\max_{y\in\mathcal{R}[X;-I]} \frac{\|[A,B]y\|}{\|y\|} < \min_{y\in\mathcal{R}[I;X^\top]} \frac{\|[A,B]y\|}{\|y\|}. \tag{2.2}$$

Proof. Denote the SVD of $[A,B]$ with the descending order of singular values by

$$[A,B]=\left(\sum_{i=1}^{n} u_i\sigma_i v_i^\top\right)+\left(\sum_{i=n+1}^{n+d} u_i\sigma_i v_i^\top\right):=\left[\widehat{A},\widehat{B}\right]+\left[\widetilde{A},\widetilde{B}\right]:=U_1\Sigma_1V_1^\top+U_2\Sigma_2V_2^\top.$$

Then, X is the true solution to (1.1) if and only if $[\widehat{A},\widehat{B}][X^\top;-I]^\top=\widehat{A}X-\widehat{B}=\mathbf{0}$ [41]. Denote $[I;X^\top](I+XX^\top)^{-\frac{1}{2}}=T_1$ and $[X;-I](I+X^\top X)^{-\frac{1}{2}}=T_2$. It is easy to verify that $T_1^\top T_1=I, T_2^\top T_2=I$, $T_1^\top T_2=\mathbf{0}$, $T_1T_1^\top+T_2T_2^\top=I$, and $[T_1,T_2]$ is orthogonal. Write

$$\begin{cases}\overline{A}=\left(A+BX^\top\right)\left(I+XX^\top\right)^{-\frac{1}{2}}:=[A,B]T_1,\\ \overline{R}=(AX-B)\left(I+X^\top X\right)^{-\frac{1}{2}}:=[A,B]T_2.\end{cases}$$

Therefore, (2.1) is exactly $\overline{A}^\top\overline{R}=\mathbf{0}$ and (2.2) is equivalent to $\sigma_{\max}(\overline{R})<\sigma_{\min}(\overline{A})$. Thus, we shall alternatively prove that

$$X \text{ is the true mTLS solution} \Longleftrightarrow \overline{A}^\top\overline{R}=\mathbf{0} \text{ and } \sigma_{\max}(\overline{R})<\sigma_{\min}(\overline{A}).$$

For the necessity, if X is the mTLS solution, then $\widehat{A}X-\widehat{B}=\mathbf{0}$, indicating that $V_1^\top T_2=\mathbf{0}$, i.e., $\mathcal{R}(T_2)\perp\mathcal{R}(V_1)$. Since $\dim(V_1)+\dim(T_2)=n+d$, we have $\mathcal{R}(T_2)=\mathcal{R}(V_1)^\perp$ and thus $\mathcal{R}(T_1)=\mathcal{R}(T_2)^\perp=(\mathcal{R}(V_1)^\perp)^\perp=\mathcal{R}(V_2)^\perp$. We immediately get $V_2^\top T_1=\mathbf{0}$. Then, we have

$$\begin{aligned}\overline{A}^\top\overline{R}&=\left([A,B]T_1\right)^\top\left([A,B]T_2\right)\\ &=\left(U_1\Sigma_1V_1^\top T_1\right)^\top\left(U_2\Sigma_2V_2^\top T_2\right)=T_1^\top V_1\Sigma_1\left(U_1^\top U_2\right)\Sigma_2V_2^\top T_2=\mathbf{0}.\end{aligned}$$

Notice that $\overline{A}=U_1\Sigma_1(T_1^\top V_1)^\top$ and $\overline{R}=U_2\Sigma_2(T_2^\top V_2)^\top$ are essentially the SVDs of $\overline{A}$ and $\overline{R}$, respectively. Therefore, $\sigma_{\max}(\overline{R})=\sigma_{\max}(\Sigma_2)<\sigma_{\max}(\Sigma_1)=\sigma_{\min}(\overline{A})$.

For the sufficiency, we follow the same notation as above and write the SVDs as $\overline{A}=U_1'\Sigma_1'V_1'^\top$ and $\overline{R}=U_2'\Sigma_2'V_2'^\top$. Then, we have

$$[A,B]=[A,B]\left(T_1T_1^\top+T_2T_2^\top\right)=\overline{A}T_1^\top+\overline{R}T_2^\top=[U_1',U_2']\begin{bmatrix}\Sigma_1' & \mathbf{0}\\ \mathbf{0} & \Sigma_2'\end{bmatrix}[T_1V_1',T_2V_2']^\top. \tag{2.3}$$

Since $\overline{A}^\top\overline{R}=\mathbf{0}$, we have $U_1'^\top U_2'=\mathbf{0}$. $[T_1V_1',T_2V_2']=[T_1,T_2]\left[\begin{smallmatrix}V_1' & \mathbf{0}\\ \mathbf{0} & V_2'\end{smallmatrix}\right]$ is orthogonal. Furthermore, $\sigma_{\max}(\overline{R})<\sigma_{\min}(\overline{A})$ implies that $\sigma_{\max}(\Sigma_2')<\sigma_{\min}(\Sigma_1')$. Therefore, (2.3) is indeed the SVD of $[A,B]$ and we have

$$\left[\widehat{A},\widehat{B}\right]T_2=U_1'\Sigma_1'V_1'^\top\left(T_1^\top T_2\right)=\mathbf{0}\Longrightarrow\widehat{A}X-\widehat{B}=\mathbf{0}.$$

This indicates that X is the true mTLS solution. We complete the whole proof. ∎

Remark 2.2. Equation (2.1) implies that the true mTLS solution X is a minimizer of $f(X) = \|(B - AX)(I + X^\top X)^{-\frac{1}{2}}\|_{\mathrm{F}}$, which is consistent with a TLS case [2, 13]. We note that this necessary condition has also been observed in earlier work [41, 44]. In addition, (2.1) is equivalent to the following relation by some algebra:

$$A^\top (B - AX) = -X \left(X^\top X + I\right)^{-1} (B - AX)^\top (B - AX). \tag{2.4}$$

We emphasize (2.1) which the true solution X should satisfy because it enables us to characterize perturbations in the mTLS system in an elegant way. Specifically, given a computed solution $\widetilde{X}$, we can define the relaxed feasible set of perturbations $[E, F]$ that make $\widetilde{X}$ the true solution to a nearby mTLS system as

$$\mathcal{F}_{\mathrm{mTLS+}} = \left\{[E,F] : \left((A+E) + (B+F)\widetilde{X}^\top\right)^\top \left((B+F) - (A+E)\widetilde{X}\right) = \mathbf{0}\right\}.$$

The relaxation comes from ignoring (2.2) in Theorem 2.1. Fortunately, it is usable for computing the backward error in practice. We give some justifications here.

Proposition 2.3 (Asymptotical accuracy of the relaxed feasible set). *Suppose the optimal perturbations coming from $\mathcal{F}_{\mathrm{mTLS+}}(\widetilde{X})$ are E^* and F^* with the weighted backward error given by $\|[E^*, \theta F^*]\|_{\mathrm{F}}$. Then, as $\widetilde{X} \to X_{\mathrm{mTLS}}$, $\widetilde{X}$ is bound to be the true mTLS solution to $[A + E^*, B + F^*]$.*

Proof. By the definition of $\mathcal{F}_{\mathrm{mTLS+}}$, $[A + E^*, B + F^*]$ satisfies (2.1) with respect to $\widetilde{X}$. Then by Theorem 2.1, we only need to prove that when $\widetilde{X} \to X_{\mathrm{mTLS}}$,

$$\max_{y \in \mathcal{R}\left([\widetilde{X}; -I]\right)} \frac{\|[A + E^*, B + F^*]y\|}{\|y\|} \le \min_{y \in \mathcal{R}\left([I; \widetilde{X}^\top]\right)} \frac{\|[A + E^*, B + F^*]y\|}{\|y\|}. \tag{2.5}$$

Notice that $\|X\| \le \|X\|_{\mathrm{F}}$ holds for any matrix X and thus it suffices to show that

$$\max_{y \in \mathcal{R}\left([\widetilde{X}; -I]\right)} \frac{\|[A, B]y\|}{\|y\|} + \|[E^*, F^*]\|_{\mathrm{F}} \le \min_{y \in \mathcal{R}\left([I; \widetilde{X}^\top]\right)} \frac{\|[A, B]y\|}{\|y\|} - \|[E^*, F^*]\|_{\mathrm{F}},$$

which can be rearranged to

$$2\|[E^*, F^*]\|_{\mathrm{F}} \le \min_{y \in \mathcal{R}\left([I; \widetilde{X}^\top]\right)} \frac{\|[A, B]y\|}{\|y\|} - \max_{y \in \mathcal{R}\left([\widetilde{X}; -I]\right)} \frac{\|[A, B]y\|}{\|y\|}.$$

The functions above are continuous with respect to $\widetilde{X}$. As $\widetilde{X} \to X_{\mathrm{mTLS}}$, the left-hand side tends to 0 because the explicit expression presented later in Theorem 2.8 indicates that

$$\|[E^*, F^*]\|_{\mathrm{F}}^2 = \|\widetilde{R}\|_{\mathrm{F}}^2 + \min_Y \mathrm{tr}\left[Y^\dagger \left(\widetilde{A}\widetilde{A}^\top - \widetilde{R}\widetilde{R}^\top\right) Y\right],$$

where $\widetilde{A} = (A + B\widetilde{X}^\top)(I + \theta^{-2}\widetilde{X}\widetilde{X}^\top)^{-\frac{1}{2}}$ and $\widetilde{R} = \theta(B - A\widetilde{X})(I + \theta^2\widetilde{X}^\top\widetilde{X})^{-\frac{1}{2}}$. By taking $Y = \widetilde{R}$, we have $\|[E^*, F^*]\|_{\mathrm{F}} \le \|\widetilde{A}^\top \widetilde{R}\|_{\mathrm{F}}$, which tends to 0 by (2.2) as $\widetilde{X} \to X_{\mathrm{mTLS}}$. Meanwhile, the right-hand side tends to $\sigma_n([A, B]) - \sigma_{n+1}([A, B]) > 0$ and thus the inequality holds. ∎

Therefore, we will directly use $F_{\mathrm{mTLS+}}$ or relevant relaxed sets for later analysis. Simplifying $\mathcal{F}_{\mathrm{mTLS+}}$ is our next goal.

2.2. Characterization of perturbations in mTLS problems. A similar problem on characterizing perturbations to the LS systems with multiple right-hand sides has been studied in [37]. Given $A \in \mathbb{R}^{m\times n}, B \in \mathbb{R}^{m\times d}$, and the solution $\widehat{X}_{\rm LS} \in \mathbb{R}^{n\times d}$ with full column rank, let $\widehat{R}_{\rm LS} = B - A\widehat{X}_{\rm LS}$ and define

$$\mathcal{E}_{\rm LS} = \left\{ E \in \mathbb{R}^{m\times n} : \left\| (A+E)\widehat{X}_{\rm LS} - B \right\|_{\rm F} = \min_{X\in\mathbb{R}^{n\times d}} \|(A+E)X - B\|_{\rm F} \right\};$$

then $\mathcal{E}_{\rm LS} = \mathcal{M}_{\rm LS}$, where

$$\mathcal{M}_{\rm LS} = \left\{ -YY^\dagger A + (I - YY^\dagger)\left[\widehat{R}_{\rm LS}\widehat{X}_{\rm LS}^\dagger + Z\left(I - \widehat{X}_{\rm LS}\widehat{X}_{\rm LS}^\dagger \right) \right] : [Y,Z] \in \mathbb{R}^{m\times(n+d)} \right\}. \tag{2.6}$$

In the following, we also explore the circumstance where the right-hand side B is fixed. As discussed, $\overline{X}$ is the true solution to a nearby mTLS system $(A+E)X \approx B$ only if the additional E stems from the relaxed feasible set

$$\mathcal{E}_{\rm mTLS+} = \left\{ E \in \mathbb{R}^{m\times n} : \left((A+E) + B\overline{X}^\top \right)^\top \left(B - (A+E)\overline{X} \right) = \mathbf{0} \right\}. \tag{2.7}$$

The next theorem is vital for our analysis of the backward error.

THEOREM 2.4. *Given $A \in \mathbb{R}^{m\times n}, B \in \mathbb{R}^{m\times d}$, and $\overline{X} \in \mathbb{R}^{n\times d}$ with full column rank. Then, $\mathcal{E}_{\rm mTLS+} = \mathcal{M}_{\rm mTLS+}$, where $\mathcal{E}_{\rm mTLS+}$ is defined in* (2.7) *and*

$$\begin{aligned}\mathcal{M}_{\rm mTLS+} = \Big\{ &- YY^\dagger \left(A + B\overline{X}^\top \right) + (I - YY^\dagger)\left[\widehat{R}\overline{X}^\dagger + Z\left(I - \overline{X}\overline{X}^\dagger \right) \right] : \\ & Y \in \mathbb{R}^{m\times d}, Z \in \mathbb{R}^{m\times n} \Big\}, \quad \textit{where } \widehat{R} = B - A\overline{X}.\end{aligned} \tag{2.8}$$

Proof. We shall notice the similarity between (2.1) and the normal equation of the LS, in the form of $A^\top(B - AX) = \mathbf{0}$. Since the conditions in $\mathcal{E}_{\rm mTLS+}$ can be rewritten as

$$\left[\left(A + B\overline{X}^\top \right) + E \right]^\top \left[\left(B + B\overline{X}^\top\overline{X} - \left(A + B\overline{X}^\top + E \right)\overline{X} \right) \right] = \mathbf{0},$$

we can reduce it to a modified LS system $[\widetilde{A}, \widetilde{B}]$ that is to be perturbed by E, where $\widetilde{A} = A + B\overline{X}^\top$ and $\widetilde{B} = B + B\overline{X}^\top\overline{X}$. Therefore, by (2.6), $\mathcal{E}_{\rm mTLS+}$ is identical to the set whose element E corresponds to some $[Y,Z] \in \mathbb{R}^{m\times(n+d)}$, such that

$$\begin{aligned} E &= -YY^\dagger\widetilde{A} + (I - YY^\dagger)\left[\left(\widetilde{B} - \widetilde{A}\overline{X} \right)\overline{X}^\dagger + Z\left(I - \overline{X}\overline{X}^\dagger \right) \right] \\ &= -YY^\dagger\left(A + B\overline{X}^\top \right) + (I - YY^\dagger)\left[\left(B - A\overline{X} \right)\overline{X}^\dagger + Z\left(I - \overline{X}\overline{X}^\dagger \right) \right]. \end{aligned}$$

Namely, $\mathcal{E}_{\rm mTLS+} = \mathcal{M}_{\rm mTLS+}$. ∎

2.3. Optimal backward error for solutions with full column rank. In this section, we develop an explicit expression for the optimal weighted backward error (1.3) based on the previous results. The following lemmas are useful.

LEMMA 2.5 ([37]). *Let $H \in \mathbb{R}^{m\times m}$ be the symmetric matrix with the eigenvalues $\lambda_1 \le \lambda_2 \le \cdots \le \lambda_m$. Then,*

$$\min_{Y\in\mathbb{C}^{m\times d}} \operatorname{tr}(Y^\dagger HY) = \begin{cases} 0 & \textit{if } 0 \le \lambda_1, \\ \lambda_1 + \lambda_2 + \cdots + \lambda_s & \textit{if } \lambda_s < 0 \le \lambda_{s+1} \textit{ with } s < d, \\ \lambda_1 + \lambda_2 + \cdots + \lambda_d, & \textit{if } \lambda_d < 0. \end{cases}$$

LEMMA 2.6 ([29]). *Given $A, B \in \mathbb{R}^{m\times n}$, the solution to $(A+C)^\top (B+D) = \mathbf{0}$ can be expressed as*

$$\begin{cases} C = -YY^\dagger A + \left(I - YY^\dagger\right) M \\ D = -\left(I - YY^\dagger\right) B + YY^\dagger N \end{cases} \qquad \text{for some } Y, M, N \in \mathbb{R}^{m\times n}.$$

For the final preparation, we give the fact below, which can be directly verified and will be used repeatedly in the proof of the main result.

FACT 2.7. *Given $X \in \mathbb{R}^{n\times d}$ and $\theta \in \mathbb{R}$, we have*

$$\begin{cases} I - \left(I + \theta^2 \widetilde{X}\widetilde{X}^\top\right)^{-1} = \theta^2 \left(\widetilde{X}\widehat{X}^\top\right)\left(I + \theta^2 \widetilde{X}\widetilde{X}^\top\right)^{-1}, \\ I - \left(I + \theta^2 \widetilde{X}^\top\widetilde{X}\right)^{-1} = \theta^2 \left(\widetilde{X}^\top\widetilde{X}\right)\left(I + \theta^2 \widetilde{X}^\top\widetilde{X}\right)^{-1}, \\ I - \theta^2\left(\theta^2 I + \widetilde{X}\widetilde{X}^\top\right)^{-1} = \left(\widetilde{X}\widetilde{X}^\top\right)\left(\theta^2 I + \widetilde{X}\widetilde{X}^\top\right)^{-1}. \end{cases}$$

Given a computed mTLS approximation $\widetilde{X}$, denote $\eta_\theta(\widetilde{X})$ as defined in (1.3), which by Theorem 2.1 can be expressed as

$$\min\left\{\|[E, \theta F]\|_{\mathrm{F}} : \left((A+E) + (B+F)\widetilde{X}^\top\right)^\top \left((B+F) - (A+E)\widetilde{X}\right) = \mathbf{0}\right\}. \tag{2.9}$$

Then, we arrive at our main theorem, where we establish the weighted backward error of perturbations on both sides in the mTLS problem.

THEOREM 2.8 (Optimal weighted backward error for mTLS problems with all data perturbed). *Let $A \in \mathbb{R}^{m\times n}, B \in \mathbb{R}^{m\times d}$, and the computed approximation $\widetilde{X} \in \mathbb{R}^{n\times d}$ with full column rank. Define two matrices*

$$\begin{cases} \widetilde{A} = \left(A + B\widetilde{X}^\top\right)\left(I + \theta^{-2}\widetilde{X}\widetilde{X}^\top\right)^{-\frac{1}{2}}, \\ \widetilde{R} = \theta\left(B - A\widetilde{X}\right)\left(I + \theta^2 \widetilde{X}^\top\widetilde{X}\right)^{-\frac{1}{2}}, \end{cases} \tag{2.10}$$

where θ is the positive weighting parameter in (2.9). Write $\lambda_1^{(\theta)} \le \lambda_2^{(\theta)} \le \cdots \le \lambda_m^{(\theta)}$ as the eigenvalues of $\widetilde{A}\widetilde{A}^\top - \widetilde{R}\widetilde{R}^\top$. Then, the optimal backward error is given by

$$\eta_\theta(\widetilde{X}) = \left[\|\widetilde{R}\|_{\mathrm{F}}^2 + \lambda_*^{(\theta)}\right]^{\frac{1}{2}}, \tag{2.11}$$

where $\lambda_*^{(\theta)}$ is defined as

$$\lambda_*^{(\theta)} = \begin{cases} 0 & \text{if } 0 \le \lambda_1^{(\theta)}, \\ \lambda_1^{(\theta)} + \lambda_2^{(\theta)} + \cdots + \lambda_s^{(\theta)} & \text{if } \lambda_s^{(\theta)} < 0 \le \lambda_{s+1}^{(\theta)} \text{ with } s < d, \\ \lambda_1^{(\theta)} + \lambda_2^{(\theta)} + \cdots + \lambda_d^{(\theta)} & \text{if } \lambda_d^{(\theta)} < 0. \end{cases}$$

Proof. The condition (2.9) can be rewritten as

$$\left[\left(A + B\widetilde{X}^\top\right) + \left(E + F\widetilde{X}^\top\right)\right]^\top \left[\left(A\widetilde{X} - B\right) + \left(E\widetilde{X} - F\right)\right] = \mathbf{0}.$$

By Lemma 2.6, we know that for some Y, M, and N,

$$E + F\widetilde{X}^\top = -YY^\dagger\left(A + B\widetilde{X}^\top\right) + \left(I - YY^\dagger\right)M, \tag{2.12}$$

$$E\widetilde{X} - F = -\left(I - YY^\dagger\right)\left(A\widetilde{X} - B\right) + YY^\dagger N. \tag{2.13}$$

Right-multiplying (2.13) with $\widetilde{X}^\top$ and adding to (2.12), and right-multiplying (2.12) with $\widetilde{X}$ and subtracting (2.13) from it, we respectively obtain

$$E\left(I + \widetilde{X}\widetilde{X}^\top\right) = -YY^\dagger\left(A + B\widetilde{X}^\top - N\widetilde{X}^\top\right) + \left(I - YY^\dagger\right)\left(M + \left(B - A\widetilde{X}\right)\widetilde{X}^\top\right), \tag{2.14}$$

$$F\left(I + \widetilde{X}^\top\widetilde{X}\right) = -YY^\dagger\left(\left(A + B\widetilde{X}^\top\right)\widetilde{X} + N\right) + \left(I - YY^\dagger\right)\left(M\widetilde{X} - \left(B - A\widetilde{X}\right)\right). \tag{2.15}$$

By the orthogonality of $YY^\dagger$ and $I - YY^\dagger$, we can simplify the error as

$$\begin{aligned}
\|E\|_{\mathrm{F}}^2 + \theta^2\|F\|_{\mathrm{F}}^2 = &\underbrace{\|YY^\dagger(A + B\widetilde{X}^\top - N\widetilde{X}^\top)(I + \widetilde{X}\widetilde{X}^\top)^{-1}\|_{\mathrm{F}}^2}_{(I)} \\
&+ \underbrace{\|(I - YY^\dagger)(M - (A\widetilde{X} - B)\widetilde{X}^\top)(I + \widetilde{X}\widetilde{X}^\top)^{-1}\|_{\mathrm{F}}^2}_{(II)} \\
&+ \underbrace{\theta^2\|YY^\dagger((A + B\widetilde{X}^\top)\widetilde{X} + N)(I + \widetilde{X}^\top\widetilde{X})^{-1}\|_{\mathrm{F}}^2}_{(III)} \\
&+ \underbrace{\theta^2\|(I - YY^\dagger)(M\widetilde{X} + A\widetilde{X} - B)(I + \widetilde{X}^\top\widetilde{X})^{-1}\|_{\mathrm{F}}^2}_{(IV)}.
\end{aligned}$$

It can be verified that by taking $N^* = (1 - \theta^2)(A + B\widetilde{X}^\top)\widetilde{X}(\theta^2 I + \widetilde{X}^\top\widetilde{X})^{-1}$ and $M^* = (1 - \theta^2)(A\widetilde{X} - B)\widetilde{X}^\top(I + \theta^2\widetilde{X}\widetilde{X}^\top)^{-1}$, we have

$$\begin{aligned}
(I) + (III) \geq &\|YY^\dagger(A + B\widetilde{X}^\top - N^*\widetilde{X}^\top)(I + \widetilde{X}\widetilde{X}^\top)^{-1}\|_{\mathrm{F}}^2 \\
&+ \theta^2\|YY^\dagger((A + B\widetilde{X}^\top)\widetilde{X} + N^*)(I + \widetilde{X}^\top\widetilde{X})^{-1}\|_{\mathrm{F}}^2,
\end{aligned} \tag{2.16}$$

$$\begin{aligned}
(II) + (IV) \geq &\|(I - YY^\dagger)(M^* - (A\widetilde{X} - B)\widetilde{X}^\top)(I + \widetilde{X}\widetilde{X}^\top)^{-1}\|_{\mathrm{F}}^2 \\
&+ \theta^2\|(I - YY^\dagger)(M^*\widetilde{X} + A\widetilde{X} - B)(I + \widetilde{X}^\top\widetilde{X})^{-1}\|_{\mathrm{F}}^2.
\end{aligned} \tag{2.17}$$

At the same time, we can obtain the relations that

$$\begin{cases}
A + B\widetilde{X}^\top - N^*\widetilde{X}^\top = \theta^2\left(A + B\widetilde{X}^\top\right)\left(\theta^2 I + \widetilde{X}\widetilde{X}^\top\right)^{-1}\left(I + \widetilde{X}\widetilde{X}^\top\right), \\
\left(A + B\widetilde{X}^\top\right)\widetilde{X} + N^* = \left(A + B\widetilde{X}^\top\right)\widetilde{X}\left(\theta^2 I + \widetilde{X}^\top\widetilde{X}\right)^{-1}\left(I + \widetilde{X}^\top\widetilde{X}\right), \\
M^* - \left(A\widetilde{X} - B\right)\widetilde{X}^\top = -\theta^2\left(A\widetilde{X} - B\right)\widetilde{X}^\top\left(I + \theta^2\widetilde{X}\widetilde{X}^\top\right)^{-1}\left(I + \widetilde{X}\widetilde{X}^\top\right), \\
M^*\widetilde{X} + A\widetilde{X} - B = \left(A\widetilde{X} - B\right)\left(I + \theta^2\widetilde{X}^\top\widetilde{X}\right)^{-1}\left(I + \widetilde{X}^\top\widetilde{X}\right).
\end{cases} \tag{2.18}$$

If the constructions of N^*, M^* and more details about (2.18) are of interest, please refer to Appendix A for some mathematical tricks.

Plugging (2.18) into (2.16) and (2.17), we have

$$
\begin{aligned}
&(I)+(III)\\
&\geq \mathrm{tr}\left[Y^{\dagger}(A+B\widetilde{X}^{\top})\cdot\theta^2(\theta^2 I+\widetilde{X}\widehat{X}^{\top})^{-1}\cdot\theta^2(\theta^2 I+\widetilde{X}\widehat{X}^{\top})^{-1}(A+B\widetilde{X}^{\top})^{\top}Y\right]\\
&\quad+\theta^2\mathrm{tr}\left[Y(A+B\widetilde{X}^{\top})\cdot(\theta^2 I+\widetilde{X}\widetilde{X}^{\top})^{-1}(\widetilde{X}\widetilde{X}^{\top})(\theta^2 I+\widetilde{X}\widetilde{X}^{\top})^{-1}(A+B\widetilde{X}^{\top})^{\top}Y\right]\\
&=\mathrm{tr}\left(Y^{\dagger}\widetilde{A}\widetilde{A}^{\top}Y\right),\\
&(II)+(IV)\\
&\geq \mathrm{tr}\left[(I+\theta^2\widetilde{X}\widetilde{X}^{\top})^{-1}\cdot\theta^2\widetilde{X}(A\widetilde{X}-B)^{\top}(I-YY^{\dagger})\cdot\theta^2(A\widetilde{X}-B)\widetilde{X}^{\top}(I+\theta^2\widetilde{X}\widetilde{X}^{\top})^{-1}\right]\\
&\quad+\theta^2\mathrm{tr}\left[(I+\theta^2\widetilde{X}^{\top}\widetilde{X})^{-1}(A\widetilde{X}-B)^{\top}(I-YY^{\dagger})(A\widetilde{X}-B)(I+\theta^2\widetilde{X}^{\top}\widetilde{X})^{-1}\right]\\
&=\theta^2\mathrm{tr}\left[(I+\theta^2\widetilde{X}^{\top}\widetilde{X})^{-1}(A\widetilde{X}-B)^{\top}(I-YY^{\dagger})(A\widetilde{X}-B)\widetilde{X}^{\top}(I-(I+\theta^2\widetilde{X}^{\top}\widetilde{X})^{-1})\right]\\
&\quad+\theta^2\mathrm{tr}\left[(I+\theta^2\widetilde{X}^{\top}\widetilde{X})^{-1}(A\widetilde{X}-B)^{\top}(I-YY^{\dagger})(A\widetilde{X}-B)(I+\theta^2\widetilde{X}^{\top}\widetilde{X})^{-1}\right]\\
&=\theta^2\mathrm{tr}\left[(I+\theta^2\widetilde{X}^{\top}\widetilde{X})^{-1}(A\widetilde{X}-B)^{\top}(I-YY^{\dagger})(A\widetilde{X}-B)\right]=\|\widetilde{R}\|_{\mathrm{F}}^2-\mathrm{tr}\left(Y^{\dagger}\widetilde{R}\widetilde{R}^{\top}Y\right).
\end{aligned}
$$

We fully exploit Fact 2.7 in the above derivations. Finally, we complete the proof by applying Lemma 2.5 to minimize over Y. ∎

We remark that the optimal perturbations E^*, F^* can be immediately constructed by substituting N^*, M^* and the optimal Y^* into (2.14) and (2.15). The following two corollaries concerning the single right-hand side case and the circumstance where only A is perturbed are direct consequences of Theorem 2.8.

Corollary 2.9 (Optimal backward error of the TLS with single right-hand side). *Let $A \in \mathbb{R}^{m\times n}, b \in \mathbb{R}^m$, and the computed approximation $\widetilde{x} \in \mathbb{R}^n$. Define $\widetilde{A} = (A + b\widetilde{x}^{\top})(I + \theta^{-2}\widetilde{x}\widetilde{x}^{\top})^{-\frac{1}{2}}$ and $\widetilde{r} = \theta(1 + \theta^2\|\widetilde{x}\|^2)^{-\frac{1}{2}}(b - A\widetilde{x})$. Denote λ_θ as the smallest eigenvalue of the matrix $\widetilde{A}\widetilde{A}^{\top} - \widetilde{r}\widetilde{r}^{\top}$. Then, the optimal backward error $\tau_\theta(\widetilde{x})$ is given by*

$$\tau_\theta(\widetilde{x}) = \left[\|\widetilde{r}\|^2 + \min\{0, \lambda_\theta\}\right]^{\frac{1}{2}}. \tag{2.19}$$

Corollary 2.10 (Optimal backward error for mTLS problems with fixed right-hand sides). *Let $A \in \mathbb{R}^{m\times n}, B \in \mathbb{R}^{m\times d}$, and an approximation $\overline{X} \in \mathbb{R}^{n\times d}$ with* $\mathrm{rank}(\overline{X}) = d$. *Write $\widehat{R} = B - A\overline{X}$ as the residual, and assume no perturbation in B. Denote the eigenvalues of the matrix $(A + B\overline{X}^{\top})(A + B\overline{X}^{\top})^{\top} - \widehat{R}(\overline{X}^{\top}\overline{X})^{-1}\widehat{R}^{\top}$ as $\lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_m$. Then, the optimal backward error for A is given by*

$$\eta_B(\overline{X}) = \left[\left\|\widehat{R}\left(\overline{X}^{\top}\overline{X}\right)^{-\frac{1}{2}}\right\|_{\mathrm{F}}^2 + \lambda_*\right]^{\frac{1}{2}}, \tag{2.20}$$

where λ_* is defined as

$$\lambda_* = \begin{cases} 0 & \text{if } 0 \leq \lambda_1,\\ \lambda_1 + \lambda_2 + \cdots + \lambda_s & \text{if } \lambda_s < 0 \leq \lambda_{s+1} \text{ with } s < d,\\ \lambda_1 + \lambda_2 + \cdots + \lambda_d & \text{if } \lambda_d < 0.\end{cases}$$

Proof. It suffices to set $\theta=\infty$ in Theorem 2.8, and by Fact 2.7, we have

$$\begin{aligned}&\lim_{\theta\to\infty}\theta^2\widehat{R}\left(I+\theta^2\overline{X}^\top\overline{X}\right)^{-1}\widehat{R}^\top\\&\quad=\lim_{\theta\to\infty}\widehat{R}\left(\overline{X}^\top\overline{X}\right)^{-1}\left[I-\left(I+\theta^2\overline{X}^\top\overline{X}\right)^{-1}\right]\widehat{R}^\top=\widehat{R}\left(\overline{X}^\top\overline{X}\right)^{-1}\widehat{R}^\top,\end{aligned}$$

and thus $\lim_{\theta\to\infty}\|\theta\widehat{R}(I+\theta^2\overline{X}^\top\overline{X})^{-\frac{1}{2}}\|_{\mathrm{F}}^2=\|\widehat{R}(\overline{X}^\top\overline{X})^{-\frac{1}{2}}\|_{\mathrm{F}}^2$. ☐

For completeness, we generalize the above results to the rank-deficient case, with the complete proof given in Appendix B. While (2.1) and (2.4) are equivalent, our previous discussions primarily adopted (2.1). In this generalization, we base the proof on (2.4) to illustrate that the same line of reasoning can be developed from either formulation.

THEOREM 2.11. *Let $A\in\mathbb{R}^{m\times n},B\in\mathbb{R}^{m\times d}$, and $\widetilde{X}=(\widetilde{X}_1,\mathbf{0})\in\mathbb{R}^{m\times d}$ with $\widetilde{X}_1\in\mathbb{R}^{m\times r_1}$ be full column rank. Define the backward error for perturbations in A as*

$$\eta(\widetilde{X})=\min\left\{\|E\|_{\mathrm{F}}:\|(A+E)\widetilde{X}-B\|_{\mathrm{F}}=\min_X\|(A+E)X-B\|_{\mathrm{F}}\right\}.$$

Moreover, let $B=(B_1,B_2)$ with $B_1\in\mathbb{R}^{m\times r_1},B_2\in\mathbb{R}^{m\times r_2}$, and $\operatorname{rank}(B_2)=r_2,\widehat{R}_1=B_1-A\widetilde{X}_1$. Define $r_*=\min\{m-r_2,r_1\}$, and let $\lambda_1\le\lambda_2\le\cdots\le\lambda_m$ be the eigenvalues of the matrix $(A+B_1\widetilde{X}_1^\top)(A+B_1\widetilde{X}_1^\top)^\top-\widehat{R}_1(\widetilde{X}_1^\top\widetilde{X}_1)^{-1}\widehat{R}_1^\top$. Then, writing $P_{B_2}=B_2B_2^\dagger$ and $P_{B_2}^\perp=I-P_{B_2}$, we have

$$\eta(\widetilde{X})=\left[\|P_{B_2}A\|_{\mathrm{F}}^2+\left\|P_{B_2}^\perp\widehat{R}_1\left(\widetilde{X}_1^\top\widetilde{X}_1\right)^{-\frac{1}{2}}\right\|_{\mathrm{F}}^2+\lambda_*\right]^{\frac{1}{2}},\tag{2.21}$$

where λ_* is defined as

$$\lambda_*=\begin{cases}0 & \text{if } 0\le\lambda_1,\\ \lambda_1+\lambda_2+\cdots+\lambda_s & \text{if } \lambda_s<0\le\lambda_{s+1} \text{ with } s<r_*,\\ \lambda_1+\lambda_2+\cdots+\lambda_{r_*} & \text{if } \lambda_{r_*}<0.\end{cases}$$

3. Estimates for the optimal backward error of mTLS systems. In this section, we propose efficient estimates for Theorem 2.8 to facilitate practical applications. We give tight bounds for our estimates in the single right-hand side case and offer procedures to solve multidimensional ones. Implementation details are also discussed.

3.1. For the single right-hand side case. By Corollary 2.9, the weighted optimal backward error of a TLS for a computed solution $\widetilde{x}$ is given by

$$\tau_\theta(\widetilde{x})=\min_{E,f}\|[E,\theta f]\|_{\mathrm{F}}=\left[\|\widetilde{r}\|^2+\min\left\{0,\lambda_{\min}\left(\widetilde{A}\widetilde{A}^\top-\widetilde{r}\widetilde{r}^\top\right)\right\}\right]^{1/2},$$

where $\widetilde{A}=(A+b\widetilde{x}^\top)(I+\theta^{-2}\widetilde{x}\widetilde{x}^\top)^{-\frac{1}{2}}$ and $\widetilde{r}=\theta(1+\theta^2\|\widetilde{x}\|^2)^{-\frac{1}{2}}(b-A\widetilde{x})$. For numerical stability, similar to the considerations in the LS settings [21], $\tau_\theta(\widetilde{x})$ is better obtained by the equivalent

$$\min\left\{\|\widetilde{r}\|,\sigma_{\min}(M)\right\},\ \text{with } M=\begin{pmatrix}\widetilde{A}^\top\\ \|\widetilde{r}\|\left(I-\widetilde{r}\widetilde{r}^\dagger\right)\end{pmatrix}.$$

It is still computationally prohibitive to compute $\lambda_{\min}(M^\top M)$ directly. The following estimate guarantees an approximation up to a small constant factor.

THEOREM 3.1 (Computable estimate for a TLS). *Define*

$$\tilde{\tau}_\theta(\widetilde{x}) = \left\| \left[\widetilde{A}^\top \widetilde{A} + \|\widetilde{r}\|^2 I \right]^{-\frac{1}{2}} \widetilde{A}^\top \widetilde{r} \right\|; \tag{3.1}$$

then $\tilde{\tau}_\theta(\widetilde{x}) \le \tau_\theta(\widetilde{x}) \le \sqrt{2}\tilde{\tau}_\theta(\widetilde{x})$.

We refer the reader to [15, Theorem 3.1] for the detailed proof, as we can substitute the A therein with our $\widetilde{A}$. It is not surprising since the optimal backward errors in the LS and TLS systems mainly differ in the A and $\widetilde{A}$ as TLS takes all perturbations into account.

Furthermore, we can utilize the sketching technique to accelerate the computation of $\tau_\theta(\widetilde{x})$ without losing much accuracy. We introduce the sketching matrix and its significant property for which readers can refer to the abundant literature [19, 33, 46].

DEFINITION 3.2. *A (random) matrix* $S \in \mathbb{R}^{s\times m}$ *is called a sketching matrix for a matrix* $A \in \mathbb{R}^{m\times n}$ *with distortion* $0 < \epsilon < 1$ *if*

$$(1-\epsilon)\|Ay\| \le \|S(Ay)\| \le (1+\epsilon)\|Ay\| \quad \textit{for all } y \in \mathbb{R}^n.$$

Probabilistic sketching matrices satisfy the property with high probability.

The following lemma directly follows from the well-known results in [46].

LEMMA 3.3 (Randomized preconditioning). *Let S be a sketching matrix for a full-rank A with distortion $\epsilon \in (0,1)$. Consider the factorization in the form of $SA = QK$, where Q is orthogonal and K is square. Then,*

$$\frac{1}{1+\epsilon} \le \sigma_{\min}(AK^{-1}) \le \sigma_{\max}(AK^{-1}) \le \frac{1}{1-\epsilon}.$$

In the next theorem, we provide an efficient and relatively accurate formula $\tilde{\tau}_\theta^{sk}(\widetilde{x})$ to estimate $\tilde{\tau}_\theta(\widetilde{x})$ and thus approximate our targeted $\tau_\theta(\widetilde{x})$.

THEOREM 3.4 (Computable estimate for a TLS with sketching). *Denote* $S \in \mathbb{R}^{s\times(n+1)}$ *as the sketching matrix for* $[A, b]$ *with distortion* $\epsilon \in (0,1)$ *and* $SA = QK$ *as the factorization in Lemma* 3.3. *Following the notation before, given the computed solution* $\widetilde{x}$, *let* $\widetilde{A} = (A + b\widetilde{x}^\top)(I + \theta^{-2}\widetilde{x}\widetilde{x}^\top)^{-\frac{1}{2}}$ *and* $\widetilde{r} = \theta(1+\theta^2\|\widetilde{x}\|^2)^{-\frac{1}{2}}(b - A\widetilde{x})$. *Write*

$$\tilde{\tau}_\theta^{sk}(\widetilde{x}) = \left\| \left(\left(S\widetilde{A}\right)^\top \left(S\widetilde{A}\right) + \|\widetilde{r}\|^2 I \right)^{-\frac{1}{2}} \widetilde{A}^\top \widetilde{r} \right\|; \tag{3.2}$$

then $(1-\epsilon)\tilde{\tau}_\theta^{sk}(\widetilde{x}) \le \tau_\theta(\widetilde{x}) \le \sqrt{2}\cdot(1+\epsilon)\tilde{\tau}_\theta^{sk}(\widetilde{x})$.

Proof. This proof is essentially the same as for the LS case [12]. Write $z = \widetilde{A}^\top\widetilde{r}$ and then

$$\begin{aligned}\left(\frac{\tilde{\tau}_\theta^{sk}(\widetilde{x})}{\tilde{\tau}_\theta(\widetilde{x})}\right)^2 &= \frac{z^\top \left(\left(S\widetilde{A}\right)^\top \left(S\widetilde{A}\right) + \|\widetilde{r}\|^2 I \right)^{-1} z}{z^\top \left(\widetilde{A}^\top \widetilde{A} + \|\widetilde{r}\|^2 I\right)^{-1} z} \\ &\ge \lambda_{\min}\left(\left(\widetilde{A}^\top\widetilde{A} + \|\widetilde{r}\|^2 I\right)^{-\frac{1}{2}} \left(\left(S\widetilde{A}\right)^\top \left(S\widetilde{A}\right) + \|\widetilde{r}\|^2 I \right) \left(\widetilde{A}^\top\widetilde{A} + \|\widetilde{r}\|^2 I\right)^{-\frac{1}{2}} \right).\end{aligned}$$

Algorithm 3.1. Efficient estimate for the optimal backward error of a TLS.

Require: Matrix $A \in \mathbb{R}^{m\times n}$, right-hand side $b \in \mathbb{R}^m$, computed solution $\widetilde{x} \in \mathbb{R}^n$, weighting parameter θ

Ensure: Sketched estimate $\tilde{\tau}_\theta^{sk}(\widetilde{x})$ of the optimal backward error

1: Draw a sketching matrix $S \in \mathbb{R}^{s\times m}$ for $[A, b]$

2: $\widetilde{r} \leftarrow \theta(1+\theta^2\|\widetilde{x}\|^2)^{-\frac{1}{2}}(b - A\widetilde{x})$, $\zeta^2 \leftarrow \widetilde{r}^\top \widetilde{r}$

3: $\widetilde{A} \leftarrow (A + b\widetilde{x}^\top)(I+\theta^{-2}\widetilde{x}\widetilde{x}^\top)^{-\frac{1}{2}}$ ▷ Forming $\widetilde{A}$ efficiently by rank-one update

4: $(\widetilde{U}_{\rm sk}, \widetilde{\Sigma}_{\rm sk}, \widetilde{V}_{\rm sk}) \leftarrow \mathtt{svd}(S\widetilde{A})$ ▷ $\widetilde{\Sigma}_{sk}$ in column vector form

5: $\tilde{\tau}_\theta^{sk}(\widetilde{x}) \leftarrow \|[\frac{1}{\sigma_i^2+\zeta^2}]_{i=1}^n \odot (\widetilde{V}_{sk}^\top \widetilde{A}^\top \widetilde{r})\|$ ▷ Hadamard product $\odot$

By the definition of the minimum eigenvalue and some algebra, we have

$$\left(\frac{\tilde{\tau}_\theta^{sk}(\widetilde{x})}{\tilde{\tau}_\theta(\widetilde{x})}\right)^2 \geq \min_{\omega\neq 0} \frac{\omega^\top\left(\left(S\widetilde{A}\right)^\top\left(S\widetilde{A}\right)+\|\widetilde{r}\|^2 I\right)\omega}{\omega^\top\left(\widetilde{A}^\top\widetilde{A}+\|\widetilde{r}\|^2 I\right)\omega} = \min_{\omega\neq 0}\frac{\frac{\|S\widetilde{A}\omega\|^2}{\|\widetilde{A}\omega\|^2}+\frac{\|\widetilde{r}\|^2\|\omega\|^2}{\|\widetilde{A}\omega\|^2}}{1+\frac{\|\widetilde{r}\|^2\|\omega\|^2}{\|\widetilde{A}\omega\|^2}}$$

$$\geq \min\left\{\min_{\omega\neq 0}\frac{\|S\widetilde{A}\omega\|^2}{\|\widetilde{A}\omega\|^2}, 1\right\} \geq \min\left\{\min_{\omega\neq 0}\frac{\|K\omega\|^2}{\|\widetilde{A}\omega\|^2}, 1\right\} \geq (1-\epsilon)^2.$$

The second inequality uses the fact that for $\alpha, x > 0$, $(\alpha+x)/(1+x) \geq \min\{\alpha, 1\}$, and the last inequality uses Lemma 3.3. We can obtain the upper bound with the same idea and the fact that for $\alpha, x > 0$, $(1+x)/(\alpha+x) \leq \max\{1/\alpha, 1\}$. Finally, together with Theorem 3.1, we complete the proof. ∎

Therefore, we summarize the procedure for estimating the optimal backward error of a TLS in Algorithm 3.1 whose complexity is $\mathcal{O}(mn\log n)$. We remark that the computation of $(I+\theta^{-2}\widetilde{x}\widetilde{x}^\top)^{-\frac{1}{2}}$ in the term $\widetilde{A}$ can be done efficiently via the rank-one update theory [3] as

$$\left(I+\theta^{-2}\widetilde{x}\widetilde{x}^\top\right)^{-\frac{1}{2}} = I + \left(\frac{\theta}{\sqrt{\theta^2+\|\widetilde{x}\|^2}} - 1\right)\widetilde{x}\widetilde{x}^\dagger.$$

Implementation details. A good sketching matrix should be efficient for multiplications and have a satisfying distortion property as well. Reference [33] offers various options, while we tend to use sparse sign embeddings according to the timing experiments in [10]. A sparse embedding matrix $S \in \mathbb{R}^{s\times m}$ is in the form of

$$\mathcal{S} = \xi^{-\frac{1}{2}}[s_1, s_2, \ldots, s_m],$$

whose columns $s_i \in \mathbb{R}^s$ are independent random vectors with ξ nonzero entries placed in uniformly random positions. The nonzero entries are ± 1 with equal probability. Such S with distortion parameter $\epsilon \in (0,1)$ can be proven to be a sketching matrix for $[A, b] \in \mathbb{R}^{m\times(n+1)}$, provided that [8]

$$s \geq k_1\frac{(n+1)\log(n+1)}{\epsilon^2}, \quad \xi \geq k_2\frac{\log(n+1)}{\epsilon}, \text{ where } k_1, k_2 \text{ are constants.}$$

Following the discussions in [7, 11, 39], we take $s \approx 12n$ and $\xi = 8$ in Algorithm 3.1. Numerical experiments in section 4 illustrate that the estimate is accurate, aligning with the theoretical predictions in Theorems 3.1 and 3.4.

3.2. For the multiple right-hand sides case. For simplicity, we assume the computed solution $\widetilde{X} \in \mathbb{R}^{n\times d}(d<n)$ is of full rank in the following discussions. The rank-deficient case can be handled similarly. Recall from Theorem 2.8 that

$$\eta_\theta(\widetilde{X}) = \left[\|\widetilde{R}\|_{\mathrm{F}}^2 + \mathrm{tr}_-\left(\widetilde{A}\widetilde{A}^\top - \widetilde{R}\widetilde{R}^\top\right)\right]^{\frac{1}{2}},$$

where $\widetilde{A} = (A+\widetilde{B}\widetilde{X}^\top)(I+\theta^{-2}\widetilde{X}\widetilde{X}^\top)^{-\frac{1}{2}}$ and $\widetilde{R} = \theta(\widetilde{B}-A\widetilde{X})(I+\theta^2\widetilde{X}^\top\widetilde{X})^{-\frac{1}{2}}$. $\mathrm{tr}_-(C)$ denotes the sum of nonpositive eigenvalues of the matrix C. The following lemma is an extension to [20, Theorem 4.2] where the multidimensional least squares is considered.

LEMMA 3.5. $\eta_\theta(\widetilde{X})$ *has an equivalent formulation as*

$$\eta\left(\widetilde{A},\widetilde{R}\right) = \min\left\{\|[M,N]\|_{\mathrm{F}} : \widetilde{A}^\top N + M^\top\widetilde{R} = -\widetilde{A}^\top\widetilde{R} \text{ and } M^\top N = \mathbf{0}\right\}.$$

It is obvious that $\eta(\widetilde{A},\widetilde{R})$ has a natural lower bound $\nu_\theta(\widetilde{X})$ given by

$$\nu_\theta(\widetilde{X}) = \min\left\{\|[M,N]\|_{\mathrm{F}} : \widetilde{A}^\top N + M^\top\widetilde{R} = -\widetilde{A}^\top\widetilde{R}\right\}. \tag{3.3}$$

In fact, $\eta_\theta(\widetilde{X})$ can also be upper bounded by multiples of $\nu_\theta(\widetilde{X})$.

THEOREM 3.6. $1 \le \frac{\eta_\theta(\widetilde{X})}{\nu_\theta(\widetilde{X})} \le \sqrt{2}$.

Proof. The proof is exactly [20, Theorem 4.7] with changing notation. We present the details here for the convenience of the reader. By the definition of $\eta(\cdot,\cdot)$, we have $\eta(\widetilde{A},\widetilde{R}) = \eta([\begin{smallmatrix}\widetilde{A}\\ \mathbf{0}\end{smallmatrix}],[\begin{smallmatrix}\widetilde{R}\\ \mathbf{0}\end{smallmatrix}])$. Let (M,N) satisfy $\widetilde{A}^\top N + M^\top\widetilde{R} = -\widetilde{A}^\top\widetilde{R}$ and $\|[M,N]\|_{\mathrm{F}} = \nu_\theta(\widetilde{X})$. Specifically, by the Karush–Kuhn–Tucker (KKT) condition and some algebra, $M = \Pi_{\widetilde{R}}M, N = \Pi_{\widetilde{A}}N$, where $\Pi_{\widetilde{R}}, \Pi_{\widetilde{A}}$ are respectively the projections onto the column spaces of $\widetilde{R}$ and $\widetilde{A}$. Choose P_M, P_N such that $P_M^\top P_N = -M^\top N$. Then $([\begin{smallmatrix}M\\ P_M\end{smallmatrix}],[\begin{smallmatrix}N\\ P_N\end{smallmatrix}])$ has the relation that

$$\left(\begin{bmatrix}\widetilde{A}\\ 0\end{bmatrix} + \begin{bmatrix}M\\ P_M\end{bmatrix}\right)^\top \left(\begin{bmatrix}\widetilde{R}\\ 0\end{bmatrix} + \begin{bmatrix}N\\ P_N\end{bmatrix}\right) = \left(\widetilde{A}+M\right)^\top(\widetilde{R}+N) - P_M^\top P_N = \mathbf{0}.$$

Therefore, we have

$$\eta^2\left(\widetilde{A},\widetilde{R}\right) \le \left\|\left[\begin{bmatrix}M\\ P_M\end{bmatrix},\begin{bmatrix}N\\ P_N\end{bmatrix}\right]\right\|_{\mathrm{F}}^2 = \|[M,N]\|_{\mathrm{F}}^2 + \|[P_M,P_N]\|_{\mathrm{F}}^2. \tag{3.4}$$

Consider the smallest $\|[P_M,P_N]\|_{\mathrm{F}}^2$ among all feasible pairs (P_M,P_N), namely,

$$\min\|[P_M,P_N]\|_{\mathrm{F}}^2, \text{ s.t. } P_M^\top P_N = -M^\top N,$$

whose solution (P_M^*,P_N^*), by [35], satisfies $\|[P_M^*,P_N^*]\|_{\mathrm{F}}^2 = 2\|M^\top N\|_*$. Recall that $\|\cdot\|_*$ denotes the nuclear norm. Therefore, we have

$$\left\|M^\top N\right\|_* = \left\|M^\top\Pi_{\widetilde{R}}\Pi_{\widetilde{A}}N\right\|_* \le \left\|M^\top\Pi_{\widetilde{R}}\Pi_{\widetilde{A}}\right\|_{\mathrm{F}}\|N\|_{\mathrm{F}} \le \|M\|_{\mathrm{F}}\|N\|_{\mathrm{F}} \le \frac{1}{2}\|[M,N]\|_{\mathrm{F}}^2.$$

The first inequality is due to a generalized version of Hölder's inequality for Schatten norms [1]. The second uses the fact that $\|\Pi_{\widetilde{R}}\| = \|\Pi_{\widetilde{A}}\| = 1$. The third follows from the root mean square–geometric mean inequality. Together with (3.4), we have the upper bound and thus complete the proof. ∎

Algorithm 3.2. Efficient estimate for the optimal backward error of mTLS.

Require: Matrix $A \in \mathbb{R}^{m\times n}$, right-hand side $B \in \mathbb{R}^{m\times d}$, computed solution $\widetilde{X} \in \mathbb{R}^{n\times d}$, weighting parameter θ

Ensure: Sketched estimate $\nu_\theta^{sk}(\widetilde{X})$ of the optimal backward error

1: Draw a sketching matrix S for span$([A,B])$

2: $\overline{A} \leftarrow A + B\widetilde{X}^\top$, $R \leftarrow B - A\widetilde{X}$

3: $L_1 \leftarrow \texttt{chol}(I + \theta^{-2}\widetilde{X}\widetilde{X}^\top)$, $L_2 \leftarrow \texttt{chol}(I + \theta^2 \widetilde{X}^\top \widetilde{X})$ $\quad\triangleright$ $L_{1,2}$ is upper triangular

4: $\widetilde{A} \leftarrow L_1^{-1}\overline{A}^\top$ $\quad\triangleright$ $\widetilde{A} = (I + \theta^{-2}\widetilde{X}\widetilde{X}^\top)^{-\frac{1}{2}}(A + \widetilde{B}\widetilde{X}^\top)^\top$

5: $\widetilde{R} \leftarrow \theta L_2^{-1} R^\top$ $\quad\triangleright$ $\widetilde{R} = \theta(I + \theta\widetilde{X}^\top\widetilde{X})^{-\frac{1}{2}}(B - A\widetilde{X})^\top$

6: $(\widetilde{U}, \widetilde{\Sigma}, \widetilde{V}) \leftarrow \texttt{svd}(S\widetilde{A}^\top, \texttt{econ})$ $\quad\triangleright$ $\widetilde{\Sigma} = \mathrm{diag}\,(\sigma_i)_{i=1}^{r_{\widetilde{A}}}$ in column vector form

7: $(W, \Lambda, Z) \leftarrow \texttt{svd}(\widetilde{R}^\top, \texttt{econ})$ $\quad\triangleright$ $\Lambda = \mathrm{diag}\,(\lambda_j)_{j=1}^{r_{\widetilde{R}}}$ in column vector form

8: $\nu_\theta^{sk}(\widetilde{X})^2 \leftarrow 0$

9: **for** $j = 1$ to $r_{\widetilde{R}}$ **do**

10: $\quad\nu_\theta^{sk}(\widetilde{X})^2 \leftarrow \nu_\theta^{sk}(\widetilde{X})^2 + \lambda_j^2 \|[\frac{1}{\sigma_i^2+\lambda_j^2}]_{i=1}^{r_{\widetilde{A}}} \odot (\widetilde{V}^\top \widetilde{A}^\top w_j)\|^2$

11: **end for**

12: $\nu_\theta^{sk}(\widetilde{X}) \leftarrow \sqrt{\nu_\theta^{sk}(\widetilde{X})^2}$

Denote the compact SVD of $\widetilde{A} = \widetilde{U}\widetilde{\Sigma}\widetilde{V}^\top$ with $\widetilde{\Sigma} = \mathrm{diag}\,(\sigma_i)$ and $\widetilde{R} = W\Lambda Z^\top$ with $\Lambda = \mathrm{diag}(\lambda_i)$. Write $M = W\widehat{M}^\top\widetilde{V}^\top$ and $N = \widetilde{U}\widehat{N}Z^\top$; then $\widetilde{A}^\top N + M^\top\widetilde{R} = -\widetilde{A}^\top F$, leading to $\widetilde{\Sigma}\widehat{N} + \widehat{M}\Lambda = -\widetilde{\Sigma}(\widetilde{U}^\top W)\Lambda$. Therefore, we can rewrite the problem in (3.3) as

$$\nu_\theta(\widetilde{X}) = \min\left\{\left\|\left[\widehat{M}, \widehat{N}\right]\right\|_{\mathrm{F}} : \widetilde{\Sigma}\widehat{N} + \widehat{M}\Lambda = -\widetilde{\Sigma}\left(\widetilde{U}^\top W\right)\Lambda\right\}. \tag{3.5}$$

Assume the ranks of $\widetilde{A}$ and $\widetilde{R}$ are respectively $r_{\widetilde{A}}$ and $r_{\widetilde{R}}$, and denote the column partitions $\widetilde{U} = [\tilde{u}_i]$ and $W = [w_i]$. Then, the solution to (3.5) can be obtained by the KKT condition with some simple algebra that

$$\widehat{M}_{ij} = \frac{-\sigma_i^2\lambda_j\left(\tilde{u}_i^\top w_j\right)}{\sigma_i^2 + \lambda_j^2}, \quad \widehat{N}_{ij} = \frac{-\sigma_i\lambda_j^2\left(\tilde{u}_i^\top w_j\right)}{\sigma_i^2 + \lambda_j^2}.$$

Therefore, we can calculate that

$$\begin{aligned}\nu_\theta(\widetilde{X})^2 &= \sum_{i=1}^{r_{\widetilde{A}}}\sum_{j=1}^{r_{\widetilde{R}}} \frac{\sigma_i^2\lambda_j^2}{\sigma_i^2+\lambda_j^2}(\tilde{u}_i^\top w_j)^2 \\ &= \sum_{j=1}^{r_{\widetilde{R}}} \lambda_j^2 \left\|\left(\widetilde{\Sigma}^2 + \lambda_j^2 I\right)^{-\frac{1}{2}}\widetilde{\Sigma}\widetilde{U}^\top w_j\right\|_2^2 = \sum_{j=1}^{r_{\widetilde{R}}} \lambda_j^2 \left\|\left(\widetilde{A}^\top\widetilde{A} + \lambda_j^2 I\right)^{-\frac{1}{2}}\widetilde{A}^\top w_j\right\|^2.\end{aligned}$$

Similar to subsection 3.1, we can approximate $\nu_\theta(\widetilde{X})$ by the sketching technique with the same bound as in Theorem 3.4. Namely, by constructing a sketching matrix $S \in \mathbb{R}^{s\times m}$ for span$([A,B])$ with distortion parameter $\epsilon \in (0,1)$, we instead compute

$$\nu_\theta^{sk}(\widetilde{X})^2 = \sum_{j=1}^{r_{\widetilde{R}}} \lambda_j^2 \left\|\left((S\widetilde{A})^\top(S\widetilde{A}) + \lambda_j^2 I\right)^{-\frac{1}{2}}\widetilde{A}^\top w_j\right\|^2, \text{ with } 1-\epsilon \le \frac{\eta_\theta}{\nu_{sk}^\theta} \le \sqrt{2}(1+\epsilon).$$

The complete procedure is summarized in Algorithm 3.2. Remark that the matrices $\widetilde{A}$ and $\widetilde{R}$ arising from steps 3–5 typically exhibit numerical low-rank structure because $\widetilde{X}\widetilde{X}^\top$ is a rank-d update to the identity matrix.

4. Numerical experiments. In this section, we conduct numerical experiments to verify our theoretical results, including the estimate bounds and the time complexity, and show some applications with the optimal backward error estimates to improve iterative algorithms. All the experiments are implemented in MATLAB R2023a on a MacBook Pro with an Apple M2 Pro chip featuring a total of 10 cores and 16 GB of memory. Codes for the experiments can be found at https://github.com/JudyShan/mTLS-Efficient-Backward-Error-Estimate.

Recall that the targeted mTLS system $AX \approx B$ has $A \in \mathbb{R}^{m\times n}$, $B \in \mathbb{R}^{m\times d}$ with $m \gg n > d$.

Tightness of the estimates. First, we test the estimation bounds in Theorems 3.1, 3.4, and 3.6 for the TLS system with a single right-hand side ($d=1$) and the mTLS system ($d=10$). The weighting parameter is set to be $\theta = 1$. We generate different instances of $[A, B]$ with `[m,n]=[3000,20]`, but a range of condition numbers, by generating logarithmically spaced singular values and Haar random singular vectors. We test 200 random examples for each condition number and plot the density of the ratios. As shown in Figure 1, the accurate estimate (blue) follows the theoretical bounds of $1 \le \tau(\widetilde{x})/\tilde{\tau}(\widetilde{x}) \le \sqrt{2}$ or $1 \le \eta(\widetilde{X})/\nu(\widetilde{X}) \le \sqrt{2}$, regardless of the systems being well-conditioned or not. The sketched versions also perform well.

Efficiency of the estimates. To illustrate the efficiency of our proposed Algorithm 3.2, we fix the condition number of A to be 10^6 and set different sizes as `m=[5000,10000,20000]`, `n=linspace(20,60,5)`, and `d=[1,4,16]`. The problem setup is the same as before. We plot the runtime versus the problem sizes m, n, d, respectively, in Figure 2. It can be shown that for fixed m and d, the time complexity is roughly linear with respect to n, while the state-of-the-art method, such as the SVD-based one, could be quadratic in n.

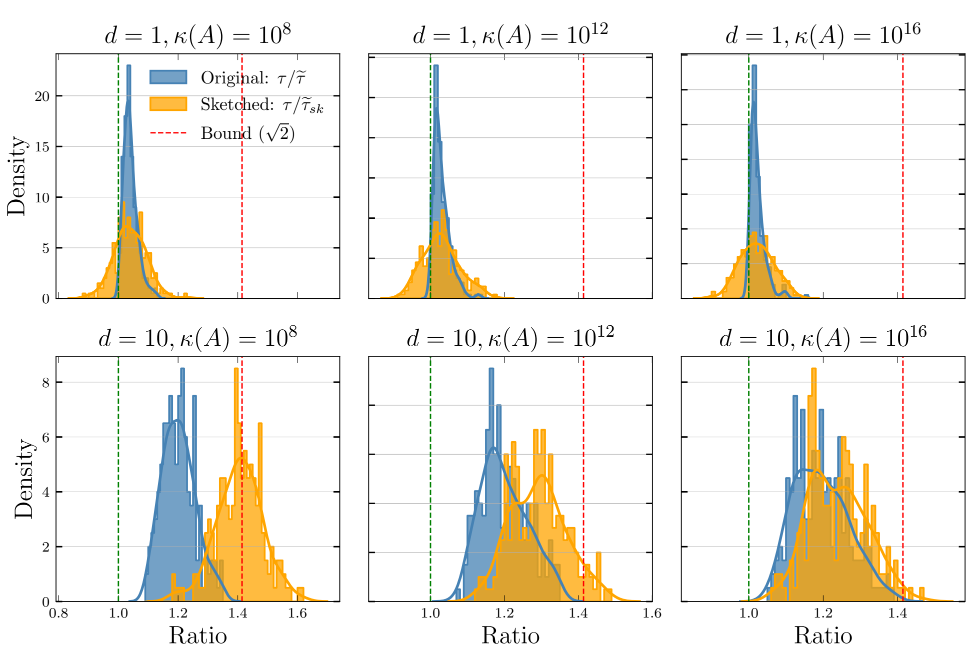

FIG. 1. *Tightness of the proposed estimations, with and without sketching techniques, for the optimal backward error of the TLS systems. (Color figure available online.)*

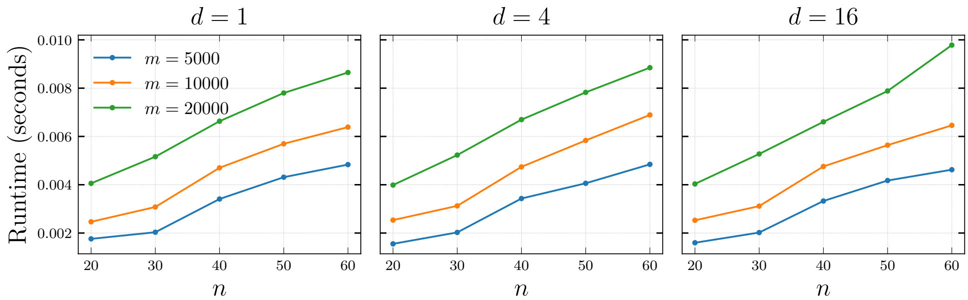

FIG. 2. *Verification for the time complexity of the sketched estimate to the optimal backward error of mTLS. Time versus n is linear, demonstrating the speedup over the classical SVD-based method.*

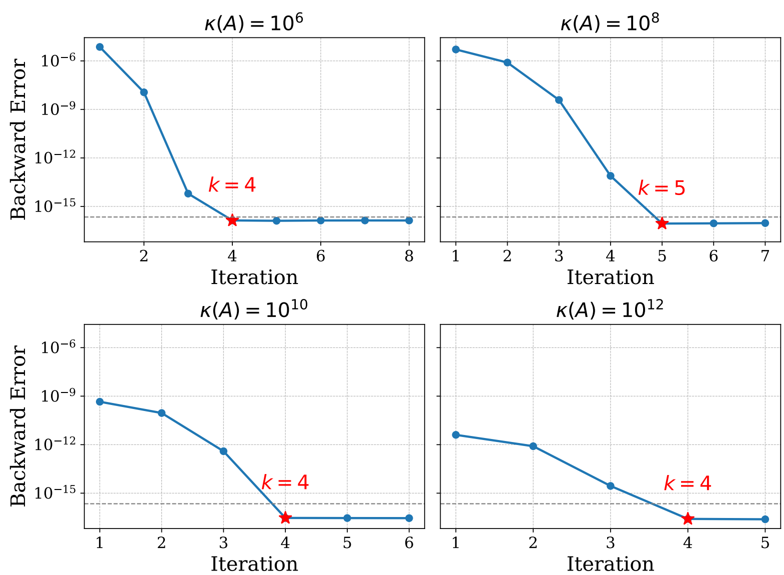

FIG. 3. *The iterative TLS algorithm with the sketched backward error estimates can stop at an early yet satisfactory time. (Color figure available online.)*

Applications of the estimates. Developing iterative algorithms itself is beyond the scope of this paper, but we aim to show that the backward error estimates, meriting the reduced complexity, can be exploited to improve the stopping criteria. We consider the classical RQI-PCGTLS algorithm proposed in [4] as an example, which uses a midstep quantity γ to control the stopping time. The logic is that mathematically, the defined γ should strictly decrease as the algorithm iterates. Therefore, when γ has an increment, it should be caused by the floating point error, and the algorithm could satisfactorily stop. However, such methods may lead to overiteration. We substitute with $\tilde{\tau}_{\theta}^{sk}(\widetilde{x})$ in subsection 3.1 and set $\tilde{\tau}_{\theta}^{sk}(\widetilde{x}) \leq u$ (the machine epsilon) to be the new stopping criteria.

In Figure 3, we test on moderate-sized problems with changing condition numbers constructed as before. The blue lines show the backward error as the original algorithm iterates, and the red stars indicate the time when the modified algorithm with the backward error estimates stops. It is illustrated that the modified algorithm can save at least 20% of the iterations while still achieving a similar backward error. We believe that such applications can be extended to better TLS solvers.

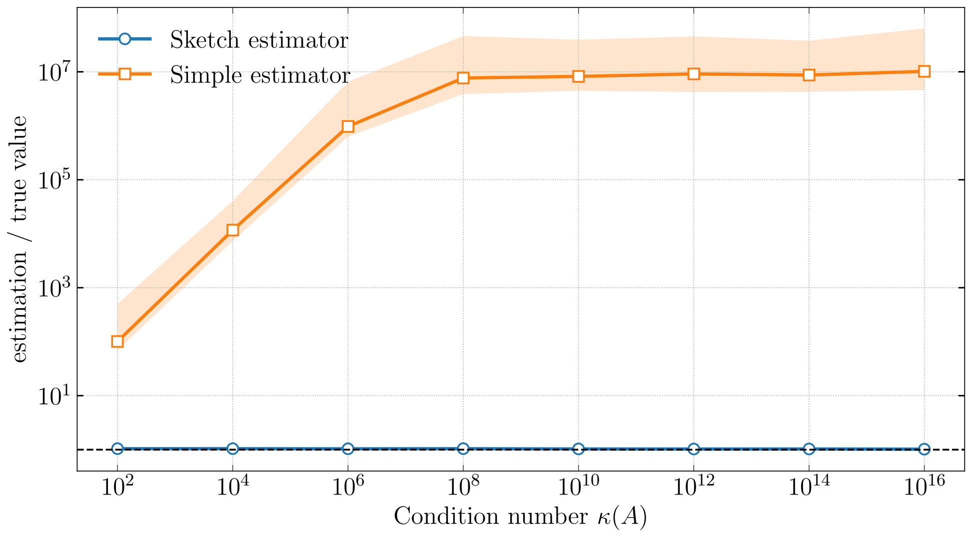

FIG. 4. *The simple estimator* $\|\widetilde{A}^\top \widetilde{r}\|/\|\widetilde{r}\|$ *may significantly overestimate the true backward error.*

The proposed estimate is more reliable. It may be tempting to try some simpler heuristics by learning from LS techniques. For example, one may consider taking $Y = \widehat{R}$ in (2.8) to obtain a natural upper bound. For a total LS $Ax \approx b$, such an idea leads to the quantity $\|\widetilde{A}^\top \widetilde{r}\|/\|\widetilde{r}\|$, where $\widetilde{A} = A + b\widetilde{x}^\top$ and $\widetilde{r} = b - A\widetilde{x}$. It is quite similar to the stopping criterion, $\|A^\top r\|/\|r\|$, for the well-known iterative LS solver LSMR, which is sometimes optimal for LS problems [38]. However, this estimator could overestimate the true backward error by a large margin. We illustrate this phenomenon in Figure 4, where we compare $\|\widetilde{A}^\top \widetilde{r}\|/\|\widetilde{r}\|$ with our sketched method across various condition numbers and a fixed problem size `[m,n]=[4000,50]`. The solid line represents the median, and the shaded area indicates the interquartile range (25%–75%). We remark that, despite the similarity in form, many of the LS results cannot be transferred trivially to TLS in general.

5. Conclusion.

In this paper, we have presented a comprehensive backward perturbation theory for the general multidimensional TLS problem, complementing existing results for the single right-hand side case and revealing intrinsic connections between the LS and TLS formulations. Backward error is critical for algorithm design and stability analysis. For practical applications, we propose efficient estimates for the explicit optimal backward error with tight bounds and offer their randomized algorithms by employing sketching techniques. Numerical experiments validate the tightness and efficiency of the estimates, showing that they can be utilized to improve the stopping criteria of iterative TLS algorithms. Future work may include developing more stable TLS algorithms based on the analysis from this paper.

Appendix A. Constructions of the optimal N^* and M^* in Theorem 2.8.

First, we point out the interchangeability between $\widetilde{X}$ or $\widetilde{X}^\top$, $I+\widetilde{X}\widetilde{X}^\top$ or $I+\widetilde{X}^\top\widetilde{X}$, and $(I+\widetilde{X}\widetilde{X}^\top)^{-1}$ or $(I+\widetilde{X}^\top\widetilde{X})^{-1}$, whenever the corresponding products are well defined. This fact will be used freely without further explanation. Recall that our first target is to minimize the sum of the following four terms with respect to M and N:

$$\begin{cases} (I) := \|YY^\dagger(A + B\widetilde{X}^\top - N\widetilde{X}^\top)(I + \widetilde{X}\widetilde{X}^\top)^{-1}\|_{\mathrm{F}}^2, \\ (II) := \|(I - YY^\dagger)(M - (A\widetilde{X} - B)\widetilde{X}^\top)(I + \widetilde{X}\widetilde{X}^\top)^{-1}\|_{\mathrm{F}}^2, \\ (III) := \theta^2\|YY^\dagger((A + B\widetilde{X}^\top)\widetilde{X} + N)(I + \widetilde{X}^\top\widetilde{X})^{-1}\|_{\mathrm{F}}^2, \\ (IV) := \theta^2\|(I - YY^\dagger)(M\widetilde{X} + A\widetilde{X} - B)(I + \widetilde{X}^\top\widetilde{X})^{-1}\|_{\mathrm{F}}^2. \end{cases}$$

It is natural to consider $(I) + (III)$ and $(II) + (IV)$ separately to seek the optimal N^* and M^*. Let us consider the special case when $\theta = 1$ to get some insights. We expand $(I) + (III)$ as

$$\begin{aligned} &(I) + (III)\Big|_{\theta=1} \\ &= \mathrm{tr}\left[(I + \widetilde{X}\widetilde{X}^\top)^{-1}(A + B\widetilde{X}^\top - N\widetilde{X}^\top)^\top YY^\dagger(A + B\widetilde{X}^\top - N\widetilde{X}^\top)(I + \widetilde{X}\widetilde{X}^\top)^{-1}\right] \\ &\quad + \mathrm{tr}\left[(I + \widetilde{X}^\top\widetilde{X})^{-1}((A + B\widetilde{X}^\top)\widetilde{X} + N)^\top YY^\dagger((A + B\widetilde{X}^\top)\widetilde{X} + N)(I + \widetilde{X}^\top\widetilde{X})^{-1}\right]. \end{aligned}$$

Notice that the cross terms between the first and second terms cancel out since

$$\begin{aligned} (I1) &= \mathrm{tr}\left[-(I + \widetilde{X}\widetilde{X}^\top)^{-1}\widetilde{X}N^\top YY^\dagger(A + B\widetilde{X}^\top)(I + \widetilde{X}\widetilde{X}^\top)^{-1}\right] \\ &= \mathrm{tr}\left[-\widetilde{X}(I + \widetilde{X}^\top\widetilde{X})^{-1}N^\top YY^\dagger(A + B\widetilde{X}^\top)(I + \widetilde{X}\widetilde{X}^\top)^{-1}\right] \\ &= -\mathrm{tr}\left[(I + \widetilde{X}^\top\widetilde{X})^{-1}N^\top YY^\dagger(A + B\widetilde{X}^\top)(I + \widetilde{X}\widetilde{X}^\top)^{-1}\widetilde{X}\right] = -(III1); \\ (I2) &= \mathrm{tr}\left[-(I + \widetilde{X}\widetilde{X}^\top)^{-1}(A + B\widetilde{X}^\top)^\top YY^\dagger N\widetilde{X}^\top(I + \widetilde{X}\widetilde{X}^\top)^{-1}\right] \\ &= -\mathrm{tr}\left[(I + \widetilde{X}\widetilde{X}^\top)^{-1}(A + B\widetilde{X}^\top)^\top YY^\dagger N(I + \widetilde{X}^\top\widetilde{X})^{-1}\widetilde{X}^\top\right] \\ &= -\mathrm{tr}\left[(I + \widetilde{X}^\top\widetilde{X})^{-1}\widetilde{X}^\top(A + B\widetilde{X}^\top)^\top YY^\dagger N(I + \widetilde{X}^\top\widetilde{X})^{-1}\right] = -(III2). \end{aligned}$$

Therefore, $(I)+(III)$ is essentially $\|YY^\dagger(A+B\widetilde{X}^\top)(I+\widetilde{X}\widetilde{X}^\top)^{-1}\|_{\mathrm{F}}^2+\|YY^\dagger N\widetilde{X}^\top(I+\widetilde{X}\widetilde{X}^\top)^{-1}\|_{\mathrm{F}}^2 + \|YY^\dagger(A + B\widetilde{X}^\top)\widetilde{X}(I + \widetilde{X}^\top\widetilde{X})^{-1}\|_{\mathrm{F}}^2 + \|YY^\dagger N\widetilde{X}(I + \widetilde{X}^\top\widetilde{X})^{-1}\|_{\mathrm{F}}^2$, the sum of several squared Frobenius norms, which can be minimized easily with $N^* = \mathbf{0}$. A similar procedure applies to $(II) + (IV)$ and thus $M^* = \mathbf{0}$. Then, it is natural to consider the general θ by the same philosophy, closely resembling the usual completion-of-squares formulation. In order to cancel out the cross terms, we reformulate certain terms in $(I) + (III)$ as

$$\begin{cases} A + B\widetilde{X}^\top - N\widetilde{X}^\top = \left(A + B\widetilde{X}^\top - N^*\widetilde{X}^\top\right) - \left((N - N^*)\widetilde{X}^\top\right), \\ \theta^2\left((A + B\widetilde{X}^\top)\widetilde{X} + N\right) = \theta^2\left[\left((A + B\widetilde{X}^\top)\widetilde{X} + N^*\right) + (N - N^*)\right], \end{cases}$$

and set

$$\theta^2\left((A + B\widetilde{X}^\top)\widetilde{X} + N^*\right) = (A + B\widetilde{X}^\top - N^*\widetilde{X}^\top)\widetilde{X} \tag{A.1}$$

to obtain, as in the proof of the main theorem, Theorem 2.8, that

$$N^* = (1 - \theta^2)(A + B\widetilde{X}^\top)\widetilde{X}(\theta^2 I + \widetilde{X}^\top\widetilde{X})^{-1}. \tag{A.2}$$

Hence, $(I) + (III)$ becomes

$$\begin{aligned} &\|YY^\dagger(A + B\widetilde{X}^\top - N^*\widetilde{X}^\top)(I + \widetilde{X}\widetilde{X}^\top)^{-1}\|_{\mathrm{F}}^2 + \|YY^\dagger(N - N^*)\widetilde{X}^\top(I + \widetilde{X}\widetilde{X}^\top)^{-1}\|_{\mathrm{F}}^2 \\ &\quad + \theta^2\|YY^\dagger((A + B\widetilde{X}^\top))\widetilde{X} + N^*(I + \widetilde{X}^\top\widetilde{X})^{-1}\|_{\mathrm{F}}^2 + \theta^2\|YY^\dagger(N - N^*)(I + \widetilde{X}^\top\widetilde{X})^{-1}\|_{\mathrm{F}}^2 \end{aligned}$$

and thus the optimal N is exactly N^*. We can do the same for $(II)+(IV)$ and set

$$M^* - (A\widetilde{X} - B)\widetilde{X}^\top = -\theta^2(M^*\widetilde{X} + A\widetilde{X} - B)\widetilde{X}^\top \tag{A.3}$$

to get

$$M^* = (1-\theta^2)(A\widetilde{X} - B)\widetilde{X}^\top(I + \theta^2\widetilde{X}\widetilde{X}^\top)^{-1}. \tag{A.4}$$

One way to form (2.18) is to substitute (A.2) and (A.4) directly and simplify the sums by Fact 2.7. We offer a simpler way here. Denoting $T = A + B\widetilde{X}^\top - N^*\widetilde{X}^\top$ and reformulating (A.1), we write

$$\begin{cases} (A + B\widetilde{X}^\top)\widetilde{X}\widetilde{X}^\top + N^*\widetilde{X}^\top = \theta^{-2}T\widetilde{X}\widetilde{X}^\top, \\ A + B\widetilde{X}^\top - N^*\widetilde{X}^\top = T. \end{cases}$$

Adding them together gives

$$A + B\widetilde{X}^\top - N^*\widetilde{X}^\top = T = (A + B\widetilde{X}^\top)(I + \widetilde{X}\widetilde{X}^\top)(I + \theta^{-2}\widetilde{X}\widetilde{X}^\top)^{-1}.$$

After rearranging, we obtain the first relation in (2.18). Substituting it back to (A.1) yields the second. Relations regarding M^* can be derived similarly.

Appendix B. Proof for Theorem 2.11. In the following, we present the proof of the optimal backward error expression (2.21) for the rank-deficient solutions to mTLS systems.

Proof. By (2.4), we have by the given notation that

$$\begin{aligned} &(A+E)^\top \left[B_1 - (A+E)\widetilde{X}_1, B_2\right] \\ &\quad = -\left[\widetilde{X}_1, \mathbf{0}\right] \begin{bmatrix} \left(\widetilde{X}_1^\top \widetilde{X}_1 + I\right)^{-1} & \mathbf{0} \\ \mathbf{0} & I \end{bmatrix} \begin{bmatrix} \left(B_1 - (A+E)\widetilde{X}_1\right)^\top \\ B_2^\top \end{bmatrix} \left[B_1 - (A+E)\widetilde{X}_1, B_2\right], \end{aligned}$$

which can be rewritten as the following two equations, namely,

$$\begin{aligned} &(A+E)^\top \left[B_1 - (A+E)\widetilde{X}_1\right] \\ &\quad = -\widetilde{X}_1 \left(\widetilde{X}_1^\top \widetilde{X}_1 + I\right)^{-1} \left[B_1 - (A+E)\widetilde{X}_1\right]^\top \left[B_1 - (A+E)\widetilde{X}_1\right] \end{aligned} \tag{B.1}$$

and

$$(A+E)^\top B_2 = -\widetilde{X}_1(\widetilde{X}_1^\top \widetilde{X}_1 + I)^{-1} \left[B_1 - (A+E)\widetilde{X}_1\right]^\top B_2. \tag{B.2}$$

Equation (B.2) can be further simplified as

$$\begin{aligned} &\left[I - \widetilde{X}_1 \left(\widetilde{X}_1^\top \widetilde{X}_1 + I\right)^{-1} \widetilde{X}_1^\top\right] (A+E)^\top B_2 \\ &\quad = -\widetilde{X}_1 \left(\widetilde{X}_1^\top \widetilde{X}_1 + I\right)^{-1} B_1^\top B_2 \\ &\quad \Longleftrightarrow \left(I + \widetilde{X}_1 \widetilde{X}_1^\top\right)^{-1} (A+E)^\top B_2 = -\widetilde{X}_1 \left(\widetilde{X}_1^\top \widetilde{X}_1 + I\right)^{-1} B_1^\top B_2 \\ &\quad \Longleftrightarrow (A+E)^\top B_2 = -\widetilde{X}_1 B_1^\top B_2, \end{aligned}$$

where the first equality is due to the Woodbury formula and the second comes from the identity that $(I+\widetilde{X}_1\widetilde{X}_1^\top)\widetilde{X}_1 = \widetilde{X}_1(I+\widetilde{X}_1^\top\widetilde{X}_1)$. Therefore, $E^\top B_2 = -(A+B_1\widetilde{X}_1^\top)^\top B_2$. By the general inverse theory, we have

$$E = -B_2B_2^\dagger\left(A+B_1\widetilde{X}_1^\top\right) + \left(I - B_2B_2^\dagger\right)K \quad \text{for any } K \in \mathbb{R}^{m\times n}.$$

Thus, we have

$$\begin{cases} A+E = -B_2B_2^\dagger B_1\widetilde{X}_1^\top + \left(I-B_2B_2^\dagger\right)K, \\ B_1-(A+E)\widetilde{X}_1 = B_1 + B_2B_2^\dagger B_1\widetilde{X}_1^\top\widetilde{X}_1 + \left(I-B_2B_2^\dagger\right)(A+K)\widetilde{X}_1. \end{cases}$$

We plug in (B.1) and by some algebra, we achieve

(B.3)

$$\begin{aligned} &(A+K)^\top\left(I-B_2B_2^\dagger\right)\left[B_1-(A+K)\widetilde{X}_1\right] \\ &\quad = -\widetilde{X}_1\left(I+\widetilde{X}_1^\top\widetilde{X}_1\right)^{-1}\left[B_1-(A+K)\widetilde{X}_1\right]^\top\left(I-B_2B_2^\dagger\right)\left[B_1-(A+K)\widetilde{X}_1\right]. \end{aligned}$$

Notice that $I-B_2B_2^\dagger$ is a projection matrix. Then, it tells us that $\widetilde{X}_1$ is the mTLS solution to the system $(I-B_2B_2^\dagger)(A+K)X \approx (I-B_2B_2^\dagger)B_1$. Denote the SVD of B_2 as $B_2 = U\left(\begin{smallmatrix}\mathbf{0}\\ \Sigma_2\end{smallmatrix}\right)V^\top$, where Σ_2 is an r_2-diagonal submatrix and $U=(U_1,U_2)$ with $U_1\in\mathbb{R}^{m\times(m-r_2)}$. Define $\widetilde{A} = U^\top A = \left(\begin{smallmatrix}\widetilde{A}_1\\ \widetilde{A}_2\end{smallmatrix}\right)$, with $\widetilde{A}_1\in\mathbb{R}^{(m-r_2)\times n}$, $\widetilde{K}=U^\top K = \left(\begin{smallmatrix}\widetilde{K}_1\\ \widetilde{K}_2\end{smallmatrix}\right)$, with $\widetilde{K}_1\in\mathbb{R}^{(m-r_2)\times n}$, and $\widetilde{B}_1 = U^\top B_1 = \left(\begin{smallmatrix}\widetilde{B}_{11}\\ \widetilde{B}_{12}\end{smallmatrix}\right)$, with $\widetilde{B}_{11}\in\mathbb{R}^{(m-r_2)\times r_1}$. Then, by multiplying $UU^\top$ to (B.3), we have

$$\begin{aligned} &\left(\widetilde{A}_1+\widetilde{K}_1\right)^\top\left(\widetilde{B}_{11}-\left(\widetilde{A}_1+\widetilde{K}_1\right)\widetilde{X}_1\right) \\ &\quad = -\widetilde{X}_1\left(I+\widetilde{X}_1^\top\widetilde{X}_1\right)^{-1}\left[\widetilde{B}_{11}-\left(\widetilde{A}_1+\widetilde{K}_1\right)\widetilde{X}_1\right]^\top\left[\widetilde{B}_{11}-\left(\widetilde{A}_1+\widetilde{K}_1\right)\widetilde{X}_1\right]. \end{aligned}$$

This is exactly the perturbation formulation for $\widetilde{A}_1^{(m-r_2)\times n}\widetilde{X}_1^{n\times r_1} \approx \widetilde{B}_{11}^{(m-r_2)\times r_1}$. By Theorem 2.4, for some $Y\in\mathbb{R}^{(m-r_2)\times r_1}$ and $Z\in\mathbb{R}^{(m-r_2)\times n}$, we have

$$\widetilde{K}_1 = -\widetilde{Y}\widetilde{Y}^\dagger\left(\widetilde{A}_1+\widetilde{B}_{11}\widetilde{X}_1^\top\right) + \left(I-\widetilde{Y}\widetilde{Y}^\dagger\right)\left[\widehat{R}_{11}\widetilde{X}_1^\dagger + Z\left(I-\widetilde{X}_1\widetilde{X}_1^\dagger\right)\right],$$

where $\widehat{R}_{11} = \widetilde{B}_{11}-\widetilde{A}_1\widetilde{X}_1 = U_1^\top\widehat{R}_1$. Thus, we denote $P_{B_2} = B_2B_2^\dagger$ and we have

$$\begin{aligned} \|E\|_{\mathrm{F}}^2 &= \left\|-B_2B_2^\dagger\left(A+B_1\widetilde{X}_1^\top\right)+\left(I-B_2B_2^\dagger\right)K\right\|_{\mathrm{F}}^2 \\ &= \left\|P_{B_2}\left(A+B_1\widetilde{X}_1^\top\right)\right\|_{\mathrm{F}}^2 + \left\|\widetilde{K}_1\right\|_{\mathrm{F}}^2, \end{aligned}$$

where the last equality is due to the orthogonality between $B_2B_2^\dagger$ and $I-B_2B_2^\dagger$ and the fact that the Frobenius norm is invariant under orthogonal transformations. Furthermore, we can also write $\|\widetilde{K}_1\|_{\mathrm{F}}^2$ by the orthogonality of $YY^\dagger, I-YY^\dagger$ and $\widetilde{X}_1\widetilde{X}_1^\dagger, I-\widetilde{X}_1\widetilde{X}_1^\dagger$ as

$$\left\|YY^\dagger\left(\widetilde{A}_1+\widetilde{B}_{11}\widetilde{X}_1^\top\right)\right\| + \left\|\left(I-\widetilde{Y}\widetilde{Y}^\dagger\right)\widehat{R}_{11}\widetilde{X}_1^\dagger\right\|_{\mathrm{F}}^2 + \left\|\left(I-\widetilde{Y}\widetilde{Y}^\dagger\right)Z\left(I-\widetilde{X}_1\widetilde{X}_1^\dagger\right)\right\|_{\mathrm{F}}^2.$$

Obviously, we should take $Z=\mathbf{0}$ to achieve the minimum. Notice that $\|\widehat{R}_{11}\widetilde{X}_1^{\dagger}\|_{\mathrm{F}}^2 = \|U_1^{\top}\widehat{R}_1\widetilde{X}_1^{\dagger}\|_{\mathrm{F}}^2 = \|(I-U_2U_2^{\top})\widehat{R}_1\widetilde{X}_1^{\dagger}\|_{\mathrm{F}}^2 = \|P_{B_2}^{\perp}\widehat{R}_1\widetilde{X}_1^{\dagger}\|_{\mathrm{F}}^2$. Therefore, we have

$$\begin{aligned}
&\min \|E\|_{\mathrm{F}}^2 \\
&\quad = \left\|P_{B_2}\left(A+B_1\widetilde{X}_1^{\top}\right)\right\|_{\mathrm{F}}^2 + \left\|\widehat{R}_{11}\widetilde{X}_1^{\dagger}\right\|_{\mathrm{F}}^2 + \min_{Y}\left\|YY^{\dagger}\left(\widetilde{A}_1+\widetilde{B}_{11}\widetilde{X}_1-\widehat{R}_{11}\widetilde{X}_1^{\dagger}\right)\right\|_{\mathrm{F}}^2 \\
&\quad = \left\|P_{B_2}\left(A+B_1\widetilde{X}_1^{\top}\right)\right\|_{\mathrm{F}}^2 + \left\|P_{B_2}^{\perp}\widehat{R}_1\widetilde{X}_1^{\dagger}\right\|_{\mathrm{F}}^2 + \min\gamma(Y),
\end{aligned}$$

with $\gamma(Y) = \mathrm{tr}\{(U_1Y)^{\dagger}[(A+B_1\widetilde{X}_1^{\top})(A+B_1\widetilde{X}_1^{\top})^{\top} - (\widehat{R}_1\widetilde{X}_1^{\dagger})(\widehat{R}_1\widetilde{X}_1^{\dagger})^{\top}]U_1Y\}$ and $\mathrm{rank}(U_1Y)=\min\{r_1, m-r_2\}$. Finally, by Lemma 2.5, we complete the proof. ∎

Reproducibility of computational results. This paper has been awarded the "SIAM Reproducibility Badge: Code and data available" as a recognition that the authors have followed reproducibility principles valued by SIMAX and the scientific computing community. Code and data that allow readers to reproduce the results in this paper are available at https://github.com/JudyShan/mTLS-Efficient-Backward-Error-Estimate.

Acknowledgments. J. Shan would like to thank Fangyan Shi for the helpful discussions during her visit to Tsinghua University and valuable suggestions on the final presentation of this work. The authors would like to thank the associate editor and two anonymous reviewers for their insightful comments and instructions.

REFERENCES

[1] K. Ball, E. A. Carlen, and E. H. Lieb, *Sharp uniform convexity and smoothness inequalities for trace norms*, Invent. Math., 115 (1994), pp. 463–482, https://doi.org/10.1007/BF01231769.

[2] A. Beck, A. Ben-Tal, and M. Teboulle, *Finding a global optimal solution for a quadratically constrained fractional quadratic problem with applications to the regularized total least squares*, SIAM J. Matrix Anal. Appl., 28 (2006), pp. 425–445, https://doi.org/10.1137/040616851.

[3] D. S. Bernstein and C. F. Van Loan, *Rational matrix functions and rank-1 updates*, SIAM J. Matrix Anal. Appl., 22 (2000), pp. 145–154, https://doi.org/10.1137/S0895479898333636.

[4] Å Björck, P. Heggernes, and P. Matstoms, *Methods for large scale total least squares problems*, SIAM J. Matrix Anal. Appl., 22 (2000), pp. 413–429, https://doi.org/10.1137/S0895479899355414.

[5] X.-W. Chang, C. C. Paige, and D. Titley-Peloquin, *Characterizing matrices that are consistent with given solutions*, SIAM J. Matrix Anal. Appl., 30 (2008), pp. 1406–1420, https://doi.org/10.1137/060675691.

[6] X.-W. Chang and D. Titley-Peloquin, *Backward perturbation analysis for scaled total least-squares problems*, Numer. Linear Algebra Appl., 16 (2009), pp. 627–648, https://doi.org/10.1002/nla.640.

[7] S. Chenakkod, M. Dereziński, X. Dong, and M. Rudelson, *Optimal embedding dimension for sparse subspace embeddings*, in Proceedings of the 56th Annual ACM Symposium on Theory of Computing, 2024, pp. 1106–1117.

[8] M. B. Cohen, *Nearly tight oblivious subspace embeddings by trace inequalities*, in Proceedings of the Twenty-Seventh Annual ACM-SIAM Symposium on Discrete Algorithms, SIAM, Philadelphia, 2016, pp. 278–287, https://doi.org/10.1137/1.9781611974331.ch21.

[9] A. J. Cox and N. J. Higham, *Backward error bounds for constrained least squares problems*, BIT, 39 (1999), pp. 210–227, https://doi.org/10.1023/A:1022385611904.

[10] Y. Dong and P.-G. Martinsson, *Simpler is better: A comparative study of randomized pivoting algorithms for CUR and interpolative decompositions*, Adv. Comput. Math., 49 (2023), 66, https://doi.org/10.1007/s10444-023-10061-z.

[11] E. N. Epperly, *Fast and forward stable randomized algorithms for linear least-squares problems*, SIAM J. Matrix Anal. Appl., 45 (2024), pp. 1782–1804, https://doi.org/10.1137/23M1616790.

[12] E. N. Epperly, M. Meier, and Y. Nakatsukasa, *Fast randomized least-squares solvers can be just as accurate and stable as classical direct solvers*, Comm. Pure Appl. Math., 79 (2026), pp. 293–339, https://doi.org/10.1002/cpa.70013.

[13] G. H. Golub and C. F. Van Loan, *An analysis of the total least squares problem*, SIAM J. Numer. Anal., 17 (1980), pp. 883–893, https://doi.org/10.1137/0717073.

[14] G. H. Golub and C. F. Van Loan, *Matrix Computations*, 4th ed., John Hopkins University Press, Baltimore, 2013.

[15] S. Gratton, P. Jiránek, and D. Titley-Peloquin, *On the accuracy of the Karlson-Waldén estimate of the backward error for linear least squares problems*, SIAM J. Matrix Anal. Appl., 33 (2012), pp. 822–836, https://doi.org/10.1137/110825467.

[16] J. F. Grcar, *Optimal Sensitivity Analysis of Linear Least Squares*, Technical Report LBNL-52434, Lawrence Berkeley National Laboratory, Berkeley, CA, 2003.

[17] J. F. Grcar, M. A. Saunders, and Z. Su, *Estimates of Optimal Backward Perturbations for Linear Least Squares Problems*, eScholarship, Lawrence Berkeley National Laboratory, 2008, retrieved from https://escholarship.org/uc/item/9mq1s46t.

[18] M. Gu, *Backward perturbation bounds for linear least squares problems*, SIAM J. Matrix Anal. Appl., 20 (1998), pp. 363–372, https://doi.org/10.1137/S0895479895296446.

[19] N. Halko, P.-G. Martinsson, and J. A. Tropp, *Finding structure with randomness: Probabilistic algorithms for constructing approximate matrix decompositions*, SIAM Rev., 53 (2011), pp. 217–288, https://doi.org/10.1137/090771806.

[20] E. Hallman, *Estimating the backward error for the least-squares problem with multiple right-hand sides*, Linear Algebra Appl., 605 (2020), pp. 227–238, https://doi.org/10.1016/j.laa.2020.07.020.

[21] N. J. Higham, *Accuracy and Stability of Numerical Algorithms*, 2nd ed., SIAM, Philadelphia, 2002, https://doi.org/10.1137/1.9780898718027.

[22] K. Hirakawa and T. W. Parks, *Image denoising using total least squares*, IEEE Trans. Image Process., 15 (2006), pp. 2730–2742, https://doi.org/10.1109/TIP.2006.877352.

[23] I. Hnětynková, M. Plešinger, and D. M. Sima, *Solvability of the core problem with multiple right-hand sides in the TLS sense*, SIAM J. Matrix Anal. Appl., 37 (2016), pp. 861–876, https://doi.org/10.1137/15M1028339.

[24] I. Hnětynková, M. Plešinger, D. M. Sima, Z. Strakoš, and S. Van Huffel, *The total least squares problem in $AX \approx B$: A new classification with the relationship to the classical works*, SIAM J. Matrix Anal. Appl., 32 (2011), pp. 748–770, https://doi.org/10.1137/100813348.

[25] I. Hnětynková, M. Plešinger, and Z. Strakoš, *The core problem within a linear approximation problem $AX \approx B$ with multiple right-hand sides*, SIAM J. Matrix Anal. Appl., 34 (2013), pp. 917–931, https://doi.org/10.1137/120884237.

[26] I. Hnětynková, M. Plešinger, and J. Žáková, *TLS formulation and core reduction for problems with structured right-hand sides*, Linear Algebra Appl., 555 (2018), pp. 241–265, https://doi.org/10.1016/j.laa.2018.06.016.

[27] I. Hnětynková, M. Plešinger, and J. Žáková, *Krylov subspace approach to core problems within multilinear approximation problems: A unifying framework*, SIAM J. Matrix Anal. Appl., 44 (2023), pp. 53–79, https://doi.org/10.1137/21M1462155.

[28] R. A. Horn and C. R. Johnson, *Topics in Matrix Analysis*, Cambridge University Press, Cambridge, 1994.

[29] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, Cambridge, 2012.

[30] R. Karlson and B. Waldén, *Estimation of optimal backward perturbation bounds for the linear least squares problem*, BIT, 37 (1997), pp. 862–869, https://doi.org/10.1007/BF02510356.

[31] Q. Liu, Z. Jia, and Y. Wei, *Multidimensional total least squares problem with linear equality constraints*, SIAM J. Matrix Anal. Appl., 43 (2022), pp. 124–150, https://doi.org/10.1137/21M1400420.

[32] I. Markovsky, J. C. Willems, S. Van Huffel, B. De Moor, and R. Pintelon, *Application of structured total least squares for system identification and model reduction*, IEEE Trans. Automat. Control, 50 (2005), pp. 1490–1500, https://doi.org/10.1109/TAC.2005.856643.

[33] P.-G. Martinsson and J. A. Tropp, *Randomized numerical linear algebra: Foundations and algorithms*, Acta Numer., 29 (2020), pp. 403–572, https://doi.org/10.1017/S0962492920000021.

[34] C. C. Paige and Z. Strakoš, *Scaled total least squares fundamentals*, Numer. Math., 91 (2002), pp. 117–146, https://doi.org/10.1007/s002110100314.

[35] B. Recht, M. Fazel, and P. A. Parrilo, *Guaranteed minimum-rank solutions of linear matrix equations via nuclear norm minimization*, SIAM Rev., 52 (2010), pp. 471–501, https://doi.org/10.1137/070697835.

[36] J. Shan and Y. Wei, *Optimal backward error of a total least squares and its randomized algorithms*, SIAM J. Matrix Anal. Appl., 46 (2025), pp. 2116–2139, https://doi.org/10.1137/25M1749657.

[37] J.-g. Sun, *Optimal backward perturbation bounds for the linear least-squares problem with multiple right-hand sides*, IMA J. Numer. Anal., 16 (1996), pp. 1–11, https://doi.org/10.1093/imanum/16.1.1.

[38] J.-g. Sun, *On optimal backward perturbation bounds for the linear least squares problem*, BIT, 37 (1997), pp. 179–188, https://doi.org/10.1007/BF02510179.

[39] J. A. Tropp, A. Yurtsever, M. Udell, and V. Cevher, *Streaming low-rank matrix approximation with an application to scientific simulation*, SIAM J. Sci. Comput., 41 (2019), pp. A2430–A2463, https://doi.org/10.1137/18M1201068.

[40] S. Van Huffel and J. Vandewalle, *Algebraic connections between the least squares and total least squares problems*, Numer. Math., 55 (1989), pp. 431–449, https://doi.org/10.1007/BF01396047.

[41] S. Van Huffel and J. Vandewalle, *The Total Least Squares Problem: Computational Aspects and Analysis*, Frontiers Appl. Math. 9, SIAM, Philadelphia, 1991, https://doi.org/10.1137/1.9781611971002.

[42] S. Vanhuffel, H. Chen, C. Decanniere, and P. Vanhecke, *Algorithm for time-domain NMR data fitting based on total least squares*, J. Magn. Reson. A, 110 (1994), pp. 228–237, https://doi.org/10.1006/jmra.1994.1209.

[43] B. Waldén, R. Karlson, and J.-G. Sun, *Optimal backward perturbation bounds for the linear least squares problem*, Numer. Linear Algebra Appl., 2 (1995), pp. 271–286, https://doi.org/10.1002/nla.1680020308.

[44] M. Wei, *Algebraic relations between the total least squares and least squares problems with more than one solution*, Numer. Math., 62 (1992), pp. 123–148, https://doi.org/10.1007/BF01396223.

[45] Y. Wei, P. Stanimirovic, and M. Petkovic, *Numerical and Symbolic Computations of Generalized Inverses*, World Scientific, Hackensack, NJ, 2018.

[46] F. Woolfe, E. Liberty, V. Rokhlin, and M. Tygert, *A fast randomized algorithm for the approximation of matrices*, Appl. Comput. Harmon. Anal., 25 (2008), pp. 335–366, https://doi.org/10.1016/j.acha.2007.12.002.

[47] S.-J. Yu and Y.-F. Jing, *Possible reduced structure of the core problem within the total least square problem*, SIAM J. Matrix Anal. Appl., 46 (2025), pp. 1616–1639, https://doi.org/10.1137/24M1715209.

[48] B. Zheng, L. Meng, and Y. Wei, *Condition numbers of the multidimensional total least squares problem*, SIAM J. Matrix Anal. Appl., 38 (2017), pp. 924–948, https://doi.org/10.1137/15M1053815.